

Online Targeted Subsampling for Stochastic Gradient-MCMC in Hidden Markov Models

Diego Luca Gonzalez Gauss

Enrico Yao-Bate

Stephen Yang

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1 Introduction

Stochastic-gradient MCMC methods such as Stochastic Gradient Langevin Dynamics (SGLD) were introduced to make Bayesian posterior sampling feasible when full-data likelihood evaluations are too expensive [17]. The idea is to replace the exact posterior gradient by a cheaper stochastic estimate, add Langevin noise, and use the resulting iterates for approximate posterior inference. This is especially appealing for long sequential models, where even one exact likelihood or gradient computation may require a pass through the entire time series. The issue however is that the standard minibatch argument assumes independent observations. In a hidden Markov model (HMM), which is a natural model for time series with persistent latent regimes [3, 15], the marginal likelihood couples each observation to its neighbors through the latent state path. As such, a single observation is not an independent likelihood summand, and minibatching isolated time points gives the wrong local object for SG-MCMC.

Ma, Foti, and Fox (2017) address this difficulty by developing an SG-MCMC framework for HMMs based on marginal likelihood gradients and buffered subsequences [10]. Instead of sampling individual observations, their method samples local time blocks. A buffer around each sampled block is used for message passing, while only the center block contributes to the stochastic gradient. This makes it possible to approximate full-sequence HMM gradients without running a full forward-backward pass at every iteration. Thus the remaining subsampling choice is how to choose the sampled blocks.

The most straightforward choice is uniform block sampling, which treats all blocks as equally useful for estimating the gradient. In HMM applications, however, the inferential target is often a regime that is persistent but not well represented throughout the sequence: a rare latent state, a transition period, or an extreme event. In these cases, most blocks correspond to common-state behavior, while a smaller number of blocks contain rare states, transitions, or extreme observations that contribute disproportionately to particular emission or transition parameters. If those blocks are undersampled, as they are likely to be under a uniform sampler, the stochastic gradient can have high variance in the coordinates whose accurate estimation depends on rare-regime information.

Ou, Young, Sen, and Dunson (2024) propose targeted stochastic-gradient MCMC for HMMs, or TASS, to address this issue [13]. Instead of using a single block distribution for all the gradient

components, TASS uses component-specific sampling weights so that blocks informative for rare-state parameters are sampled more often when those parameters are updated. In practice, TASS constructs these weights by clustering the observations to obtain proxy state labels, using those labels to estimate which blocks are relevant to which parameter components, and then correcting for the resulting non-uniform sampling inside SG-MCMC.

This gives a practical way to target rare-state information, but the resulting weights are static. They are computed once from proxy labels and a pilot parameter value, then reused throughout the run. This can be a problem because the relevant block importances are functions of the current parameter value: as SGLD moves away from the pilot, the blocks that are most informative for a given emission or transition parameter may also change. A proposal that was well targeted near initialization can therefore become stale later in the chain.

To address this, we propose Online TASS in Section 3, with the core idea being to treat block importance as a quantity to be learned over the course of the run. When a sampled block gradient is computed, Online TASS records how large that block’s contribution was for the relevant parameter group, uses a discounted regression model to predict future block importances from observable block features, and converts those predictions into updated sampling probabilities. We support this method with a finite-run gradient-estimation analysis in Section 4 showing that Static TASS pays a chi-square staleness penalty when its fixed weights differ from the current oracle weights, while Online TASS is near-oracle when its learned importance predictions are accurate. Finally, in Section 5, we study this mechanism in two settings: a synthetic changepoint HMM, where rare-state block importances can be made stale in a controlled way, and a real-data benchmark based on 5-minute BTC log realized variance, where the goal is to test adaptive targeted subsampling on a genuine sequential volatility dataset.

2 Background

We’re broadly motivated by the idea that underlying regimes can be rare, disproportionately informed by different blocks, and have this apportionment of information evolve dynamically over time. To work towards a method that accommodates for these facts, we first begin with a baseline latent regime model (that we’ve seen before!).

2.1 Hidden Markov models

A hidden Markov model consists of a latent Markov chain $z_0, z_1, \dots, z_T$ on a finite state space $\{1, \dots, K\}$ and observations $y_1, \dots, y_T$ whose conditional distribution depends on the current latent state. We write A for the transition matrix and ϕ_k for the emission parameter of state k . The joint density factorizes as

$$p_\theta(z_{0:T}, y_{1:T}) = p(z_0) \prod_{t=1}^T p_A(z_t | z_{t-1}) p_{\phi_{z_t}}(y_t | z_t),$$

where $\theta = (A, \phi_1, \dots, \phi_K)$. In our experiments the emissions are Gaussian, but the sampling problem below only uses the fact that the latent states can be marginalized and the marginal likelihood differentiated with respect to a continuous parameterization of θ .

The standard computational tool for this marginalization is forward–backward message passing. These recursions allow one to compute the HMM likelihood, smoothed state probabilities, and pairwise transition probabilities without summing explicitly over all K^{T+1} latent trajectories [1, 3]. Given a prior $\pi(\theta)$, the marginal posterior is

$$p(\theta | y_{1:T}) \propto \pi(\theta) p_\theta(y_{1:T}),$$

where $p_\theta(y_{1:T})$ is the latent-state-marginal likelihood. Gradient-based samplers such as Langevin or Hamiltonian methods can then target the posterior over θ using gradients of

$$\log p(\theta \mid y_{1:T}) = \log \pi(\theta) + \log p_\theta(y_{1:T}) + \text{constant}.$$

The computational issue is that evaluating this gradient exactly requires message passing through the full sequence. This motivates stochastic-gradient approximations when T is large.

2.2 Buffered SG-MCMC for HMMs

To obtain a stochastic-gradient estimator, we follow the buffered-subsequence construction of Ma, Foti, and Fox (2017). Partition the sequence into N central blocks

$$C_n = \{a_n, \dots, b_n\}, \quad n = 1, \dots, N,$$

and let

$$C_n^B = \{a_n - B, \dots, b_n + B\} \cap \{1, \dots, T\}$$

be the buffered window around C_n . Message passing is run on C_n^B , but only the central block C_n is used to define the gradient contribution. Thus the buffer is used to approximate the outside information entering the local smoothing probabilities, while the gradient contribution itself is localized to the block being sampled. More concretely, let

$$\tilde{\gamma}_{t,k}^{(n)}(\theta) \approx \Pr_\theta(z_t = k \mid y_{1:T}), \quad \tilde{\xi}_{t,jk}^{(n)}(\theta) \approx \Pr_\theta(z_{t-1} = j, z_t = k \mid y_{1:T})$$

denote the one-state and two-state smoothing probabilities obtained by running forward-backward only on the buffered window C_n^B . For $t \in C_n$, these local marginals approximate the full-sequence smoothing probabilities that appear in the exact HMM score. The buffered block-gradient contribution is then

$$g_n^B(\theta) = \sum_{t \in C_n} \left[\sum_{k=1}^K \tilde{\gamma}_{t,k}^{(n)}(\theta) \nabla_\theta \log p_\theta(y_t \mid z_t = k) + \sum_{j,k=1}^K \tilde{\xi}_{t,jk}^{(n)}(\theta) \nabla_\theta \log p_\theta(z_t = k \mid z_{t-1} = j) \right],$$

with the transition term interpreted for transitions whose endpoint lies in the central block. In implementation the gradient is taken with respect to the unconstrained parameterization of the emission and transition parameters, but the decomposition is the same: emission-score terms are weighted by local state probabilities, and transition-score terms are weighted by local pairwise state probabilities.

The corresponding full buffered-gradient target is

$$G^B(\theta) = \nabla \log \pi(\theta) + \sum_{n=1}^N g_n^B(\theta).$$

This is the object estimated by the stochastic gradients in the rest of the paper. It differs from the exact full-sequence posterior gradient because the full-sequence messages have been replaced by buffered local messages. We take the block construction and buffer length as fixed, and focus on how to sample the block terms $g_n^B(\theta)$.

Given a block distribution $p = (p_1, \dots, p_N)$ with $p_n > 0$, and iid draws $I_1, \dots, I_M \sim p$, define

$$\hat{G}_p^B(\theta) = \nabla \log \pi(\theta) + \frac{1}{M} \sum_{m=1}^M \frac{g_{I_m}^B(\theta)}{p_{I_m}}.$$

Then, conditional on the data, θ , and the fixed block construction,

$$\mathbb{E}[\widehat{G}_p^B(\theta)] = G^B(\theta).$$

Thus the block distribution p changes the variance of the stochastic gradient estimator, not its target. Uniform block sampling corresponds to $p_n = 1/N$; TASS-type methods replace this with non-uniform block probabilities designed to reduce the variance of the same inverse-probability estimator.

2.3 Static TASS

TASS improves on uniform block sampling by choosing non-uniform, component-specific block probabilities. To see the principle, suppose for the moment that we care about a single scalar gradient coordinate i . Write $g_{n,i}^B(\theta)$ for the i th coordinate of the buffered block gradient. For one sampled block, the second moment of the inverse-probability estimator is

$$\sum_{n=1}^N \frac{\{g_{n,i}^B(\theta)\}^2}{p_n}.$$

Minimizing this quantity over probability vectors p gives

$$p_{n,i}^*(\theta) = \frac{|g_{n,i}^B(\theta)|}{\sum_{m=1}^N |g_{m,i}^B(\theta)|}.$$

Thus the variance-minimizing rule samples blocks in proportion to their absolute contribution to the coordinate being estimated. This is the basic importance-sampling calculation behind TASS.

The ideal probabilities $p_{n,i}^*(\theta)$ are not directly usable. They depend on the current parameter value and on block-gradient quantities that are expensive to compute for every block at every iteration. In HMMs, they also depend on information about the latent state sequence. Static TASS makes this rule practical by replacing the ideal quantities with a fixed proxy. It first (K -means) clusters the observations to obtain approximate state labels, then uses those proxy labels and a pilot parameter value θ_0 to compute approximate component-wise block-gradient magnitudes. These are normalized into sampling probabilities (where we can e.g. add a uniform floor)

$$p_{n,i}^{\text{static}} = (1 - \lambda)\widetilde{p}_{n,i} + \frac{\lambda}{N}, \quad \lambda \in (0, 1),$$

so that every block remains sampleable and inverse-probability weights cannot become infinite.

So the above method ameliorates the fact that blocks can have disproportionate information about the parameter components they inform. Indeed, this is discussed in the context of rare states by Ou, Sen, Young, and Dunson, including with real world data such as solar flares and stuck pipes in oil drilling operations [18]. However, we have not yet accommodated the fact that what informs these regimes may itself be dynamic, and as such our subsampling scheme may want to evolve with what is more informative for predicting the latent state.

This issue is familiar in the regime-switching literature. Classical Markov-switching models treat the latent regime as a discrete Markov chain with regime-dependent observation laws [19, 20], but subsequent work allows transition probabilities to depend on time-varying information variables [21, 22]. Related constructions in input-output HMMs and recurrent switching linear dynamical systems likewise allow observed covariates or continuous latent states to affect the evolution of the discrete regime [23, 24]. So we have some dynamic-regime models allowing latent state evidence to vary, and TASS addressing rare-state imbalance with a static proxy; but, it remains to bridge these in this setting for a subsampling distribution that adapts when blocks informative to components change online (we also want to retain some parameter awareness, perhaps).

3 Our method: Online TASS

We recover from the existing gap in methods by proposing online TASS, which is designed as the following:

3.1 Parameter groups

Online Group-Specific TASS is a practical online analogue of component-specific TASS. Rather than fitting a separate sampler for every scalar coordinate, we partition coordinates into a small number of interpretable groups,

$$\{1, \dots, d\} = \mathcal{G}_1 \cup \dots \cup \mathcal{G}_R,$$

and let P_r select the subvector corresponding to group r . For the $K = 2$ Gaussian HMM above, our default groups are

$$\mathcal{G}_1 = \{m, \delta\} \quad \text{emission location/separation,}$$

$$\mathcal{G}_2 = \{\ell_1, \ell_2\} \quad \text{emission variance,}$$

$$\mathcal{G}_3 = \{\eta_1, \eta_2\} \quad \text{transition logits.}$$

Taking each group to be a singleton coordinate recovers fully component-specific online TASS. Taking $R = 1$ recovers online shared TASS.

3.2 Feature model for block importance

For each block C_n , compute a feature vector $w_n \in \mathbb{R}^p$ from the observations in and around the block. At SGLD iteration t , the ideal group-specific target is

$$a_{n,r}(\theta_t) = \|P_r g_n(\theta_t)\|_2.$$

Because these magnitudes are positive and can vary across orders of magnitude, we learn them on the log-squared scale:

$$f_{n,t}^{(r)} = \log(a_{n,r}(\theta_t)^2 + \varepsilon_r).$$

The online model is

$$f_{n,t}^{(r)} \approx w_n^\top \beta_t^{(r)}.$$

For each group r , we fit $\beta_t^{(r)}$ by discounted ridge regression using the sampled blocks observed so far:

$$\hat{\beta}_t^{(r)} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{s < t} \rho^{t-s} \sum_{m=1}^{M_r} \left(f_{I_{s,m},s}^{(r)} - w_{I_{s,m}}^\top \beta \right)^2 + \tau \|\beta\|_2^2,$$

where $\rho \in (0, 1)$ discounts old feedback and $\tau > 0$ regularizes early iterations. This recursion is cheap because all groups share the same block features, and one evaluated block gradient gives all subvectors $P_r g_n(\theta_t)$.

3.3 Sampling rule

The fitted coefficient vector defines predicted group-specific importance

$$\hat{a}_{n,t}^{(r)} = \exp \left\{ \frac{1}{2} w_n^\top \hat{\beta}_t^{(r)} \right\}.$$

We convert these scores to a sampling distribution with a uniform floor:

$$p_{n,t}^{(r)} = (1 - \lambda) \frac{\hat{a}_{n,t}^{(r)}}{\sum_{j=1}^N \hat{a}_{j,t}^{(r)}} + \frac{\lambda}{N}, \quad \lambda \in (0, 1).$$

The floor prevents tiny sampling probabilities, stabilizes inverse-probability weights, and ensures every block remains observable to the online learner.

For group r , draw blocks

$$I_{t,1}^{(r)}, \dots, I_{t,M_r}^{(r)} \stackrel{\text{iid}}{\sim} p_t^{(r)}.$$

The group-gradient estimator is

$$\hat{G}_{t,r} = P_r \nabla \log \pi(\theta_t) + \frac{1}{M_r} \sum_{m=1}^{M_r} \frac{P_r g_{I_{t,m}^{(r)}}(\theta_t)}{p_{I_{t,m}^{(r)},t}^{(r)}}.$$

The full gradient estimator concatenates the group estimates:

$$\hat{G}_t = \sum_{r=1}^R P_r^\top \hat{G}_{t,r}.$$

In implementation, the block sets drawn for different groups can be unioned, each unique buffered block gradient can be computed once, and the resulting subvectors can be reused for all relevant group estimates and regression updates.

3.4 Algorithm

Online Group-Specific TASS. Given blocks $C_1, \dots, C_N$, features $w_1, \dots, w_N$, parameter groups $\mathcal{G}_1, \dots, \mathcal{G}_R$, warmup length T_w , discount ρ , ridge penalty τ , floor λ , and step sizes η_t :

1. Initialize θ_0 , and initialize one ridge state $(S_0^{(r)}, b_0^{(r)})$ per group with $S_0^{(r)} = 0$, $b_0^{(r)} = 0$.
2. For $t = 0, 1, \dots, T_{\text{iter}} - 1$:
 - (a) For each group r , set $p_{n,t}^{(r)} = 1/N$ during warmup. After warmup, compute

$$z_{n,t}^{(r)} = \frac{1}{2} w_n^\top \hat{\beta}_t^{(r)},$$

stabilize the logits by subtracting $\max_n z_{n,t}^{(r)}$, and set

$$p_{n,t}^{(r)} = (1 - \lambda) \text{softmax}(z_{1,t}^{(r)}, \dots, z_{N,t}^{(r)})_n + \lambda/N.$$

- (b) Draw M_r block indices independently from $p_t^{(r)}$ for each group r .

- (c) Compute buffered block gradients for the union of sampled blocks.
- (d) Form each $\widehat{G}_{t,r}$ by inverse-probability reweighting and concatenate to obtain $\widehat{G}_t$.
- (e) Perform the SGLD update

$$\theta_{t+1} = \theta_t + \frac{\eta_t}{2} \widehat{G}_t + \sqrt{\eta_t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, I_d).$$

- (f) For each sampled block and group, compute feedback

$$f^{(r)} = \log \left(\|P_r g_I(\theta_t)\|_2^2 + \varepsilon_r \right),$$

and update the discounted ridge sufficient statistics

$$\begin{aligned} S_{t+1}^{(r)} &= \rho S_t^{(r)} + \sum_{m=1}^{M_r} w_{I_{t,m}^{(r)}} w_{I_{t,m}^{(r)}}^\top, \\ b_{t+1}^{(r)} &= \rho b_t^{(r)} + \sum_{m=1}^{M_r} w_{I_{t,m}^{(r)}} f_{I_{t,m}^{(r)},t}^{(r)}, \\ \widehat{\beta}_{t+1}^{(r)} &= (S_{t+1}^{(r)} + \tau I_p)^{-1} b_{t+1}^{(r)}. \end{aligned}$$

3.5 (Sub)sampler comparisons

We compare the following methods (for details about how these are actually implemented, check implementation details).

Uniform SG-MCMC

Static TASS

Online Shared TASS One online regression and one sampling distribution target $\|g_n(\theta_t)\|_2$ for the full gradient. This is a natural ablation but can wash out rare-state coordinates.

Online Group-Specific TASS Our main method: one online regression and sampling rule per parameter group.

Online The earlier ablated online but without initializing from a K -means.

Online Group-Specific The earlier online with a sampling rule per parameter group, but without initializing from K -means.

Oracle Infeasible benchmark using all current block gradients to set $q_{r,\theta}^*(n) \propto \|P_r g_n(\theta)\|_2$.

4 Theory

This section states the results supporting the method. Proofs are deferred to Appendix B except where the argument is immediate. The theory studies the stochastic-gradient estimator conditional on a fixed buffered block construction. It does not claim an end-to-end posterior convergence theorem for adaptive SGLD. This is deliberate: finite-step SGLD bias, HMM label geometry, buffer approximation error, and gradient clipping are separate issues. Our theoretical target is the variance of the stochastic gradient estimator.

4.1 Why gradient-estimation error is a meaningful target

Our experiments focus on gradient RMSE rather than claiming full posterior recovery. This is not merely a convenience: for Langevin-based samplers, stochastic-gradient error directly controls the discrepancy between one step of the implementable algorithm and one step of the corresponding full-gradient Langevin algorithm. Welling and Teh’s SGLD argument also separates the injected Langevin noise, with variance of order η_t , from stochastic-gradient noise, whose contribution enters the update at order η_t^2 [?]. Thus, reducing gradient variance is the local variance-reduction problem underlying scalable SG-MCMC. The next proposition makes this precise for our buffered-gradient target.

Let

$$K_t^{\text{full}}(\theta, \cdot)$$

denote the one-step Gaussian transition of full-gradient SGLD using the full buffered gradient $G(\theta)$,

$$\theta^+ = \theta + \frac{\eta_t}{2}G(\theta) + \sqrt{\eta_t}\xi, \quad \xi \sim \mathcal{N}(0, I_d),$$

and let $K_t^{\text{sg}}(\theta, \cdot)$ denote the transition using an unbiased stochastic gradient $\widehat{G}_t(\theta)$ and the same injected Gaussian noise.

Proposition 4.1 (Gradient RMSE controls finite-run Langevin perturbation). *Assume that, conditional on the past $\mathcal{F}_t$ and the current state θ_t , the stochastic gradient satisfies*

$$\mathbb{E}[\widehat{G}_t(\theta_t) \mid \mathcal{F}_t] = G(\theta_t), \quad \mathbb{E}[\|\widehat{G}_t(\theta_t) - G(\theta_t)\|_2^2 \mid \mathcal{F}_t] \leq \sigma_t^2.$$

Then the one-step transition discrepancy satisfies

$$W_2^2(K_t^{\text{sg}}(\theta_t, \cdot), K_t^{\text{full}}(\theta_t, \cdot)) \leq \frac{\eta_t^2}{4} \sigma_t^2.$$

Moreover, suppose G is L -Lipschitz and couple an approximate chain (θ_t) and a full-gradient buffered ULA chain $(\bar{\theta}_t)$ using the same Gaussian noises, with $\theta_0 = \bar{\theta}_0$. Then

$$\mathbb{E} \|\theta_T - \bar{\theta}_T\|_2^2 \leq \sum_{s=0}^{T-1} \frac{\eta_s^2}{4} \mathbb{E} \sigma_s^2 \prod_{u=s+1}^{T-1} \left(1 + \frac{L\eta_u}{2}\right)^2.$$

In particular, if $\eta_t \leq \eta$ and $\sigma_t^2 \leq \sigma^2$ for $t < T$, then

$$\mathbb{E} \|\theta_T - \bar{\theta}_T\|_2^2 \leq \frac{\eta^2 \sigma^2}{4} \sum_{s=0}^{T-1} \left(1 + \frac{L\eta}{2}\right)^{2(T-1-s)}.$$

This result justifies the experimental design. Lower gradient RMSE implies a smaller one-step Wasserstein perturbation from the full-gradient Langevin kernel and, under a Lipschitz full-gradient field, a smaller coupled finite-horizon deviation from the full-gradient buffered chain. It does not assert that any particular finite-step SGLD chain has converged to the exact posterior; convergence diagnostics and posterior-chain comparisons remain empirical checks. Instead, it explains why gradient RMSE is the right first-order object for evaluating targeted subsampling.

4.2 Group-wise stochastic-gradient objective

Fix a parameter group r and suppress r temporarily. Let

$$h_n(\theta) = P_r g_n(\theta), \quad a_n(\theta) = \|h_n(\theta)\|_2,$$

and define

$$H(\theta) = \sum_{n=1}^N h_n(\theta).$$

For a block distribution p , define the group-wise second-moment objective

$$Q_{r,\theta}(p) = \sum_{n=1}^N \frac{a_n(\theta)^2}{p_n}.$$

The group-gradient estimator based on M_r draws from p has mean-squared error

$$\mathbb{E} \left\| \widehat{H}_p(\theta) - H(\theta) \right\|_2^2 = \frac{1}{M_r} \left\{ Q_{r,\theta}(p) - \|H(\theta)\|_2^2 \right\}.$$

Thus minimizing group-gradient MSE is equivalent to minimizing $Q_{r,\theta}(p)$.

Proposition 4.2 (Conditional unbiasedness of Online Group-Specific TASS). *Assume the groups $\mathcal{G}_1, \dots, \mathcal{G}_R$ form a partition of the coordinates and P_r is the coordinate projection onto group r . At iteration t , condition on $\mathcal{F}_t$, which contains θ_t , all regression states, and hence all sampling probabilities $p_t^{(r)}$. If the group block indices are sampled independently according to $p_t^{(r)}$, then*

$$\mathbb{E}[\widehat{G}_t \mid \mathcal{F}_t] = G(\theta_t).$$

Thus the online adaptation changes the conditional variance of the gradient estimator, not its conditional target.

Theorem 4.3 (Group-specific oracle and chi-square identity). *Fix a group r and parameter value θ . Let*

$$A_r(\theta) = \sum_{n=1}^N a_{n,r}(\theta), \quad q_{r,\theta}^*(n) = \frac{a_{n,r}(\theta)}{A_r(\theta)}.$$

Then $q_{r,\theta}^$ minimizes $Q_{r,\theta}(p)$ over all block distributions with positive coordinates. Moreover,*

$$Q_{r,\theta}(p) = A_r(\theta)^2 \{1 + \chi^2(q_{r,\theta}^* \| p)\}.$$

The theorem says that every method is judged by how close its block distribution is to the current group-specific oracle. The chi-square divergence is not an arbitrary diagnostic; it is the exact multiplicative excess second moment.

4.3 Static TASS staleness

Let $q_{0,r}^{\text{static}}$ denote the fixed group- r distribution produced by Static TASS near the pilot parameter value θ_0 . At time t , the current oracle is q_{r,θ_t}^* , not $q_{0,r}^{\text{static}}$. Theorem 4.3 immediately gives the static staleness identity:

$$Q_{r,\theta_t}(q_{0,r}^{\text{static}}) = A_r(\theta_t)^2 \{1 + \chi^2(q_{r,\theta_t}^* \| q_{0,r}^{\text{static}})\}.$$

The term

$$D_{t,r}^{\text{static}} = \chi^2(q_{r,\theta_t}^* \| q_{0,r}^{\text{static}})$$

is the staleness penalty. It is zero only when the frozen Static TASS distribution matches the current oracle. It grows when the blocks that matter at θ_t differ from the blocks selected by the pilot distribution.

Corollary 4.4 (Static TASS staleness). *For group r at time t ,*

$$Q_{r,\theta_t}(q_{0,r}^{\text{static}}) = Q_{r,\theta_t}(q_{r,\theta_t}^*) \{1 + D_{t,r}^{\text{static}}\}.$$

Consequently, Static TASS is near-oracle for group r if and only if its frozen distribution is close to the current oracle in chi-square divergence.

4.4 Online near-oracle guarantee

Online Group-Specific TASS predicts the log-regularized group-gradient energy

$$m_{n,t}^{(r)} = \log(a_{n,r}(\theta_t)^2 + \varepsilon_r).$$

Let $\hat{m}_{n,t}^{(r)} = w_n^\top \hat{\beta}_t^{(r)}$ be the fitted score. The next theorem says that if these scores are uniformly accurate, then the induced block distribution is near-oracle for group-gradient MSE.

Theorem 4.5 (Near-oracle online weights from score accuracy). *Fix group r and time t . Suppose*

$$\max_{1 \leq n \leq N} |\hat{m}_{n,t}^{(r)} - m_{n,t}^{(r)}| \leq \eta_{t,r}.$$

Let $p_{t,r}^{\text{online}}$ be the Online Group-Specific TASS distribution with uniform floor λ . If $A_r(\theta_t) = \sum_n a_{n,r}(\theta_t) > 0$, then

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \alpha_{t,r} Q_{r,\theta_t}(q_{r,\theta_t}^*),$$

where

$$\alpha_{t,r} = \frac{e^{\eta_{t,r}}}{1 - \lambda} \left(1 + \frac{N\sqrt{\varepsilon_r}}{A_r(\theta_t)} \right).$$

The multiplicative factor $\alpha_{t,r}$ is the online learning penalty. It is close to one when the regression prediction error $\eta_{t,r}$ is small, the uniform floor λ is modest, and the log regularization ε_r is negligible compared with the total current group-gradient magnitude.

4.5 When online group-specific TASS beats Static TASS

Combining the staleness identity and the online near-oracle theorem yields the main comparison.

Corollary 4.6 (Online beats Static TASS for one group). *At time t and group r , Online Group-Specific TASS has smaller group-gradient second moment than Static TASS whenever*

$$\alpha_{t,r} < 1 + D_{t,r}^{\text{static}},$$

that is, whenever

$$\frac{e^{\eta_{t,r}}}{1 - \lambda} \left(1 + \frac{N\sqrt{\varepsilon_r}}{A_r(\theta_t)} \right) < 1 + \chi^2(q_{r,\theta_t}^* \| q_{0,r}^{\text{static}}).$$

In words: Online Group-Specific TASS improves on Static TASS when the online prediction penalty is smaller than the static staleness penalty. This is the central theoretical message of the paper.

For an aggregate comparison across groups, choose nonnegative reporting weights $\omega_1, \dots, \omega_R$ and define

$$\mathcal{Q}_t(\{p_r\}_{r=1}^R) = \sum_{r=1}^R \omega_r Q_{r,\theta_t}(p_r).$$

Let

$$D_t^{\text{static}} = \frac{\sum_r \omega_r A_r(\theta_t)^2 D_{t,r}^{\text{static}}}{\sum_r \omega_r A_r(\theta_t)^2},$$

and

$$\bar{\alpha}_t = \frac{\sum_r \omega_r A_r(\theta_t)^2 \alpha_{t,r}}{\sum_r \omega_r A_r(\theta_t)^2}.$$

Then online beats Static TASS in the aggregate group-wise objective if

$$\bar{\alpha}_t < 1 + D_t^{\text{static}}.$$

This form is useful empirically because all terms can be estimated in the gradient-RMSE experiments where all block gradients are computed.

4.6 Why group-specific online sampling rather than shared online sampling?

Online shared TASS uses one distribution for all parameter groups, usually targeting $\|g_n(\theta_t)\|_2$. This is computationally simple but can reintroduce the problem that motivated Static TASS: groups associated with rare or stress states may be overwhelmed by common-state gradient coordinates.

The following result formalizes why group-specific sampling is preferable for the group-wise MSE objective.

Theorem 4.7 (Group-specific oracle dominates shared oracle). *Fix θ and group weights $\omega_r \geq 0$. Let*

$$a_{n,r} = \|P_r g_n(\theta)\|_2.$$

The best group-specific oracle value is

$$\mathcal{Q}_{\text{group}}^*(\theta) = \sum_{r=1}^R \omega_r \left(\sum_{n=1}^N a_{n,r} \right)^2.$$

The best shared one-distribution oracle value for the same group-wise objective is

$$\mathcal{Q}_{\text{shared}}^*(\theta) = \left(\sum_{n=1}^N \sqrt{\sum_{r=1}^R \omega_r a_{n,r}^2} \right)^2.$$

Then

$$\mathcal{Q}_{\text{group}}^*(\theta) \leq \mathcal{Q}_{\text{shared}}^*(\theta).$$

The inequality is strict unless the vectors

$$(\sqrt{\omega_1} a_{n,1}, \dots, \sqrt{\omega_R} a_{n,R})$$

are all positively collinear across blocks with nonzero importance.

This theorem says that shared sampling is a constrained special case. It is adequate when all groups care about the same blocks, but it is suboptimal when different parameter groups are informed by different parts of the time series. This is precisely the rare-state setting: high-volatility emissions and transition parameters can be driven by stress blocks, while common-state parameters are driven by ordinary blocks.

Corollary 4.8 (When online group-specific beats online shared). *Let p_t^{shared} be any single shared online sampling distribution used for all groups, and let $p_{t,r}^{\text{online}}$ be the Online Group-Specific TASS distributions. If*

$$\sum_{r=1}^R \omega_r \alpha_{t,r} A_r(\theta_t)^2 < \sum_{r=1}^R \omega_r Q_{r,\theta_t}(p_t^{\text{shared}}),$$

then Online Group-Specific TASS has smaller aggregate group-wise second moment than Online Shared TASS. In particular, even against the best possible shared distribution, Online Group-Specific TASS is guaranteed to win whenever

$$\bar{\alpha}_t < \frac{Q_{\text{shared}}^*(\theta_t)}{Q_{\text{group}}^*(\theta_t)},$$

where

$$\bar{\alpha}_t = \frac{\sum_r \omega_r \alpha_{t,r} A_r(\theta_t)^2}{\sum_r \omega_r A_r(\theta_t)^2}.$$

The ratio on the right is at least one by Theorem 4.7; it is larger than one when different groups are driven by different blocks.

The interpretation is direct. Online shared TASS can be competitive when all groups care about the same blocks. Online Group-Specific TASS is designed for the opposite regime: rare-state or stress-state parameters are informed by blocks that may not dominate the full-gradient norm. The empirical group-wise RMSE plots should therefore report not only whether the online learner predicts well, but also whether the shared-oracle gap is appreciable.

5 Experimental design

5.1 Changepoint staleness synthetic task

We now describe the synthetic task used to evaluate the subsampling schemes. This was originally actually meant to be an extension of the synthetic task in your paper (pretty much sweeping on possible rare state 3-state HMMs), and we ended up implementing this in a few possible ways (to variable success). The purpose of this experiment is essentially to isolate a failure mode that is natural for static targeted subsampling, where a sampling rule may be accurate for the pilot parameter value at which it is constructed, but become stale when the parts of the sequence that are most informative for the posterior change over time. We did this with an HMM in which the latent transition law is fixed, the rare-state structure is known from the data-generating process, and the only intervention is a single post-changepoint perturbation of the rare-state observation means. We hope to separate three effects that are otherwise difficult to distinguish: the benefit of targeting rare states, the cost of using a static pilot approximation, and the ability of an online rule to update its sampling distribution from gradient feedback.

We generate observations from a three-state Gaussian hidden Markov model. Let $z_t \in \{1, 2, 3\}$ denote the latent state and let $y_t \in \mathbb{R}$ denote the observation. Conditional on the latent trajectory,

the observations satisfy

$$y_t \mid z_t \sim \mathcal{N}(m_t(z_t), \sigma_{z_t}^2),$$

where $\sigma_k^2 = 1$ in the data-generating model. The transition matrix is held fixed within each scenario. We consider two rare-state regimes. In the one-rare-state regime, states 1 and 2 are frequent and state 3 is rare. In the two-rare-state regime, state 1 is frequent and states 2 and 3 are rare (essentially emulating the original 1 vs. 2 rare state task in your paper). These regimes are intended to distinguish the case where the sampler must allocate effort to a single rare component from the case where two rare components compete for sampling mass.

For the one-rare-state regime we use the column-stochastic transition matrix

$$A_{mone} = \begin{pmatrix} 0.990 & 0.005 & 0.495 \\ 0.005 & 0.990 & 0.495 \\ 0.005 & 0.005 & 0.010 \end{pmatrix}, \quad \mu_{mone} = (-20, 0, 20).$$

For the two-rare-state regime we use

$$A_{mtwo} = \begin{pmatrix} 0.9990 & 0.1000 & 0.1000 \\ 0.0005 & 0.9000 & 0.0000 \\ 0.0005 & 0.0000 & 0.9000 \end{pmatrix}, \quad \mu_{mtwo} = (0, -20, 20).$$

The matrices are written so that $A_{ij} = \mathbb{P}(z_t = i \mid z_{t-1} = j)$. Thus the simulation draws z_t from the z_{t-1} -th column of A .

The changepoint task modifies the observation means of the rare states after a single time point. Let $\Omega_{\text{rare}} \subset \{1, 2, 3\}$ denote the set of rare states. We set

$$t_\star = \lfloor \kappa T_{\text{train}} \rfloor, \quad \kappa = 0.5,$$

so that the changepoint occurs halfway through the training sequence. The post-changepoint observation mean is

$$m_t(k) = \mu_k + \delta d_k \mathbf{1}\{t \geq t_\star\} \mathbf{1}\{k \in \Omega_{\text{rare}}\},$$

where $\delta \geq 0$ controls the size of the perturbation. In the one-rare-state regime we use $\Omega_{\text{rare}} = \{3\}$ and $d_3 = 1$. In the two-rare-state regime we use $\Omega_{\text{rare}} = \{2, 3\}$, with $d_2 = -1$ and $d_3 = 1$. The values $\delta = 0$ recover the static baseline model, while larger values create increasing mismatch between the pilot information used by static targeted subsampling and the gradient information encountered later in the chain.

The training sequence is partitioned into contiguous blocks of length $2L + 1$. In all reported experiments we take $L = 2$, so each block contains five observations. The samplers select blocks, not individual observations. For a block n , let $g_n(\theta)$ denote the block-gradient contribution at parameter value θ . The transition and emission updates use inverse-probability weighting so that, conditional on the current sampling distribution, the stochastic gradient estimator targets the corresponding full-data gradient.

The parameter vector consists of the emission means, emission variances, and transition probabilities. We use the grouping

$$r \in \{\mu, \sigma^2, A\},$$

where μ is the vector of emission means, σ^2 is the vector of emission variances, and A is the transition matrix. The global online sampler learns a single block distribution from the scalar response $\log(\|g_n(\theta)\|^2 + \epsilon)$. The groupwise online sampler instead learns one block distribution for each group r , using the response

$$\log(\|P_r g_n(\theta)\|^2 + \epsilon),$$

where P_r denotes projection onto the coordinates in group r .

We compare six samplers: uniform block subsampling, static TASS, global online discounted ridge regression, TASS-initialised global online discounted ridge regression, groupwise online discounted ridge regression, and TASS-initialised groupwise online discounted ridge regression. The comparison is run for $\delta \in \{0, 1, 2, 4\}$ and for both the one-rare-state and two-rare-state regimes. Each method is run from the same initialisation protocol and with the same number of SGLD iterations within a given scenario.

5.2 Practical-proxy TASS RMSE sweep

The changepoint task in Section 5.1 evaluates the samplers in a regime where staleness arises from a one-time shift in the data-generating process at a known time t^* , so the chain encounters new high-information blocks because the rare-state distribution itself changes after t^* . This second synthetic task isolates the complementary failure mode in which staleness arises not from a change in the data but from a change in θ . The data-generating process is held stationary across the entire experiment; only the parameter at which we evaluate block gradients is moved away from the pilot point. The static pilot weights are constructed once at θ_0 and never recomputed, so any oracle-ratio degradation along the sweep is directly attributable to the gap between the pilot’s importance distribution and the current oracle’s.

We generate $T = 2400$ observations from a stationary three-state multivariate Gaussian HMM with observation dimension $D = 3$ and one rare state. The covariance matrices are treated as known throughout, so the unknown parameter is the concatenation of state means $\mu \in \mathbb{R}^{K \times D}$ and transition logits $\beta_A \in \mathbb{R}^{K \times K}$. The pilot θ_0 is the true data-generating parameter, and the experiment sweeps along the rare-state mean direction

$$\theta(\delta) = \theta_0 + \delta v_{\text{rare}}, \quad \delta \in \{0, 0.4, 0.8, 1.2, 1.6, 2.0, 2.4\},$$

where v_{rare} is the unit vector pushing the rare-state mean along its first observation coordinate. Because the data are not regenerated as δ changes, the same block-gradient computation is performed on the same $T = 2400$ observations at every δ , and the only thing that varies is the parameter at which the gradient is evaluated. This is what makes the experiment a clean test of the staleness mechanism rather than of any data-generating intervention.

The series is partitioned into non-overlapping central blocks of half-width $h = 6$ (block length $2h + 1 = 13$) padded with buffer $B = 4$, giving $N \approx 184$ blocks. Block features are sixteen-dimensional summaries of the observations inside each block, including per-coordinate moments, ranges, and a small number of cross-coordinate interactions; features are mean–standard-deviation normalized before being passed to the online ridge predictor. The samplers compared are uniform, static group-wise proxy TASS, static component-wise proxy TASS, plain online group-specific TASS, proxy-anchored online group-specific TASS, and two oracles. The multivariate observation space is what makes the comparison meaningful here: with $D = 3$ emission components per state, the component-wise variant maintains $K \times D = 9$ separate weight vectors against the group-wise variant’s $K = 3$, and reporting both the group oracle and the component oracle gives a concrete instantiation of the strictness side of Theorem 4.7.

At each δ both online learners run $S = 10$ adapt steps starting from the state inherited at the previous δ , drawing $M_{\text{mean}} = M_{\text{trans}} = 6$ blocks per group per step. The online ridge hyperparameters are discount $\rho = 0.99$, ridge $\tau = 50$, exploration floor λ/N with $\lambda = 0.05$, and feedback floor $\varepsilon = 10^{-8}$. The proxy-anchored variant uses anchor power $\eta = 1.0$. Gradient RMSE is computed for each sampler over $n_{\text{rep}} = 20$ replicates and reported as oracle ratios against both the group oracle and the component oracle. We also report two diagnostics that the multivariate setting

makes computable: the group-averaged chi-square staleness $\overline{D}^{\text{static}}(\delta) = \frac{1}{R} \sum_r \chi^2(q_{r,\theta(\delta)}^* \| q_{0,r}^{\text{static}})$ of the static pilot weights against the current oracle, and the group disagreement, defined as the mean chi-square distance between the group-wise and component-wise oracle weights at $\theta(\delta)$. The group disagreement is the quantity whose nonvanishing is the strictness condition in Theorem 4.7, and reporting it alongside the RMSE oracle ratios makes the theoretical comparison directly testable on this experiment.

5.3 Real data with a crypto volatility HMM

As explained during the slides (and prior), our original motivation was in exploring crypto markets. After realizing the full geometry of the problem in our mind’s eye was rather difficult, hence we decided to reduce. (We realize that our initial intention was to work with the full posterior. Indeed, we have some results that are somewhat interesting with a 2-state HMM on the full posterior, using Stan as a ground truth since it seemed to mix well from $\hat{R}$. However, truthfully, our analysis here was somewhat unclear, and anything interesting was limited to looking at state occupancies. Also, as mentioned in the slides, there wasn’t a true rare state in the sense of what we were looking for. So, since analysis is cleaner, we’ve decided ultimately to restrict a bit to looking at BTC and working with gradient RMSEs instead. Also note the perturbation we induce; this is because, truthfully, it doesn’t seem like persistent rare states are easily captured, but we still wanted to use some facts about the BTC structure, which is fairly rich and also the original intention of our project.)

The third experimental setup applies the gradient-RMSE protocol to a real intraday crypto volatility series. The data-generating process is now unknown, the stationarity assumption is approximate, and the K-means proxy labels that build the static TASS weights have to compete with a real and unknown latent regime structure.

The raw data are 1-second mid-prices for the perpetual BTC USDT futures contract on Binance, drawn from the Tardis order-book-snapshot archive for April 2026. Tardis is the industry-standard high-fidelity source for institutional crypto microstructure: it records every tick from each exchange’s native feed in a normalized format, in contrast to the aggregated 1-minute OHLCV bars from public exchange REST endpoints. This matters here because realized variance is only well-defined when the within-bar return series is observed at high frequency, so the Tardis tick data lets us treat bar-level realized variance as a low-noise observation of true intraday quadratic variation. The full extracted series covers $T_{\text{full}} = 8602$ five-minute bars (about thirty trading days); the headline sweep uses the leading $T = 1500$ bars (about five trading days) so the per-block JAX gradient JIT trace is reused efficiently, and the full-series run is a one-flag re-execution.

We construct the HMM observation y_t from raw mid-prices in four steps: compute 1-second log returns; aggregate to 5-minute bars by summing squared 1-second log returns to form realized variance rv_t (dropping any bar with $rv_t = 0$); take logs and subtract the within-hour-of-day mean to deseasonalize; and standardize to zero mean and unit variance using global moments. A $K = 2$ Gaussian HMM requires the two emission distributions to be stationary in t , and raw 5-minute BTC log-RV has both a strong intraday seasonal cycle and a slow drift in unconditional level. The hour-of-day demeaning kills the cycle and the global standardization kills the level and scale. At $T = 1500$ any longer-run dispersion drift is short enough that a rolling-window rescaling would itself be non-stationary, so we avoid it; residual non-stationarity is absorbed into the structural choice below.

The emission is univariate Gaussian, $y_t \mid z_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$, with $K = 2$ ordered states. Reference means $\mu_0 = (\mu_{\text{calm}}, \mu_{\text{stress}})$ are the sorted means of a two-component `GaussianMixture` fit to $\{y_t\}$. Reference variances are fixed at the asymmetric values $\sigma_{\text{calm}} = 0.55$ and $\sigma_{\text{stress}} = 1.6$, which

δ	Uniform	TASS	Online	Online-TASS	Online-Group	Online-TASS-Group	Best
0	-23.93	-5.48	-5.58	-5.39	-6.20	-5.27	Online-TASS-Group
1	-6.32	-5.54	-27.26	-5.88	-6.06	-25.68	TASS
2	-5.95	-5.47	-30.25	-5.98	-5.75	-6.00	TASS
4	-31.81	-6.21	-6.65	-6.51	-7.19	-6.45	TASS

Table 1: Held-out log predictive density in the one-rare-state changepoint experiment. Larger values are better. Static TASS remains the strongest method in most shifted regimes, but the online variants are competitive once the chain is run for 2000 iterations.

encode the regime asymmetry (low-variance + common vs. high-variance + rare) as a structural prior. The transition matrix is fixed at the calm-sticky / stress-leaky structure $A_0 = \begin{pmatrix} 0.94 & 0.06 \\ 0.15 & 0.85 \end{pmatrix}$ and is reparameterized into row-wise logits in the sampler. Holding σ and A fixed while sweeping μ isolates the staleness mechanism: as μ_{stress} moves, the gradient-important blocks for the emission group shift, while the transition group is approximately invariant.

The series is partitioned into non-overlapping central blocks of length $2h + 1 = 21$ ($h = 10$, ≈ 1.75 hours) padded with a buffer of $B = 6$ bars on either side, giving $N \approx 70$ blocks. Block features (per-coordinate moments and ranges of y_t inside the block, plus a small set of interaction terms) are mean-std normalized before entering the online ridge predictor. The perturbation grid is $\theta(\delta) = \theta_0 + \delta e_{\mu_{\text{stress}}}$, $\delta \in \{0, 0.5, 1.0, 1.5, 2.0\}$. The six samplers compared at each δ are uniform, static group-wise proxy TASS (frozen pilot weights at θ_0 , mixed with 10^{-3} uniform for numerical safety), static component-wise proxy TASS with budget split $M_{r,d} = \max\{1, \lfloor KM_r/(KD) \rfloor\}$, online shared TASS, online group-specific TASS (this paper), and proxy-anchored online group-specific TASS. Oracle group weights at $\theta(\delta)$ are computed exactly and provide both the RMSE benchmark and the chi-square staleness diagnostic $D_r^{\text{static}}(\delta) = \chi^2(q_{r,\theta(\delta)}^* \parallel q_{0,r}^{\text{static}})$.

Both online learners run $S = 6$ adapt steps per δ from the state inherited at the previous δ , each drawing $M_{\text{mean}} = M_{\text{trans}} = 6$ blocks per group. Online ridge hyperparameters are $\rho = 0.99$, $\tau = 50$, $\lambda = 0.05$, $\varepsilon = 10^{-8}$. At each δ we report gradient RMSE versus the exact full gradient over $n_{\text{rep}} = 50$ replicates, the oracle ratio per group, the top-decile sampling mass $\sum_{n \in \text{top}_{10\%}} p_n$, and the group-averaged staleness $\bar{D}^{\text{static}}(\delta)$. If the rare state turns out to have moderate (not vanishing) occupancy on this window, we describe it as a high-information regime rather than strictly rare; the methodological claim does not require rarity *per se*, only that some blocks carry disproportionate gradient information for some parameter groups and that this set drifts with θ .

6 Results

6.1 Changepoint staleness synthetic task

The results show that static TASS is a strong baseline in the one-rare-state setting. Across the one-rare-state runs, TASS attains the best or nearly best held-out log predictive density. For example, when $\delta = 2$, TASS obtains a held-out log predictive density of approximately -5.47 , while the best online competitors are close but worse. When $\delta = 4$, TASS again remains strongest, with a held-out log predictive density of approximately -6.21 , followed by the TASS-initialised groupwise online sampler and the TASS-initialised global online sampler. So TASS actually holds up fairly well with the changepoint, and remains informative enough to guide sampling toward the rare component, and the additional online machinery does not uniformly improve final predictive performance.

However, the proposal diagnostics show that this predictive strength is obtained by a highly concentrated proposal. In the one-rare-state runs, the ratio Q/Q_{oracle} for static TASS is several

δ	State	Uniform	TASS	Online	Online-TASS	Online-Group	Online-TASS-Group
0	1	-25.19	-19.31	-6.99	-9.63	-5.48	-27.42
0	2	-5.23	-4.90	-5.77	-4.72	-6.75	-5.19
1	1	-5.24	-4.79	-5.63	-6.56	-6.77	-6.59
1	2	-10.28	-21.06	-6.25	-5.49	-5.08	-5.43
2	1	-10.98	-22.99	-32.55	-7.58	-7.41	-5.47
2	2	-5.37	-5.34	-5.34	-6.20	-5.77	-8.62
4	1	-6.42	-6.15	-5.49	-5.53	-7.48	-7.73
4	2	-7.55	-29.86	-30.34	-9.70	-6.96	-5.93

Table 2: Held-out log predictive density in the two-rare-state changepoint experiment. Larger values are better. The table shows the main qualitative difference between the samplers: static TASS is often strong for one rare state, but can fail severely on the other; online and groupwise methods are generally more balanced across rare components.

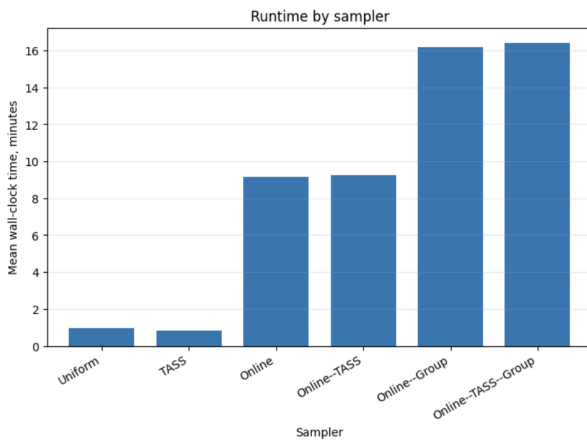


Figure 1: Wallclock time for running each method on the changepoint task.

orders of magnitude larger than for the global online and TASS-initialised online samplers. This distinction is important. The held-out likelihood indicates that the posterior chain can benefit from aggressive static targeting, but the second-moment diagnostic indicates that the corresponding importance-weighted gradient estimator is much less stable than the online alternatives. So we suppose in sum, when there is a single dominant rare component, a static proxy can remain effective even under moderate mean displacement, at the cost of highly distorted proposal weights.

In the two-rare-state setting, interestingly static TASS often concentrates on one rare component and performs poorly on the other. For instance, when $\delta = 1$, TASS gives strong predictive performance for one rare state but much worse performance for the other. Similarly when $\delta = 4$, TASS remains accurate for one rare component, while its held-out log predictive density for the other rare component deteriorates substantially. This behaviour is consistent with a static proposal that identifies one high-score proxy region and continues to allocate sampling mass according to that initial ranking.

The online methods are less aggressive but more balanced in this two-rare-state regime. They generally do not match the best state-wise performance of static TASS on the rare component that TASS identifies correctly, but they avoid the same degree of collapse on the other rare component. This is most visible in the $\delta = 1$ and $\delta = 4$ cases, where the online and groupwise online samplers produce more comparable held-out log predictive densities across the two rare states. The groupwise

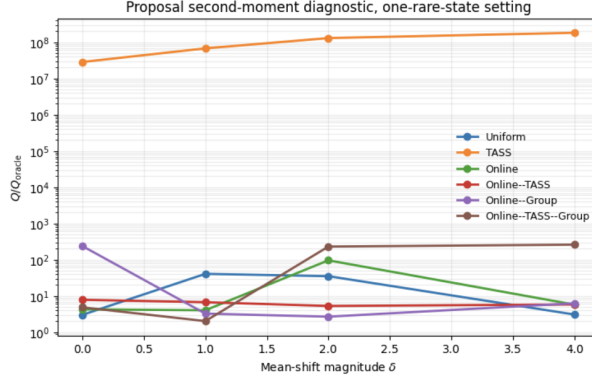


Figure 2: Q/Q_{oracle} for our different methods as a function of the mean shift at our changepoint.

variants are particularly relevant here because the transition, mean, and variance parameters need not receive their information from the same blocks. A single global proposal can average these signals, whereas a groupwise proposal can allocate sampling effort separately for emission and transition information.

Thus in aligned regimes, the static initialisation can improve the early proposal and yield competitive final predictive performance, while in misaligned regimes, the same initialisation can bias the online regression toward stale or incomplete proxy information. This is why we believe the TASS-initialised groupwise sampler is best in some cases, but not consistently so. The results suggest that static TASS initialisation is useful when the proxy labels identify the dynamically relevant rare component, and potentially harmful when several rare components compete for sampling mass.

Thus altogether static TASS is an effective but brittle sampler, where it can substantially improve predictive performance, but its proposal can be extremely concentrated and can favour one rare component over another. Online adaptation reduces this dependence on a fixed proxy by updating the proposal from observed gradient information. This makes the online methods less extreme in their second-moment diagnostics and more balanced across rare states, but it does not guarantee dominance in final held-out likelihood. The empirical role of the online methods is therefore not to replace static TASS in every regime, but to provide an adaptive correction when static proxy scores are incomplete, stale, or unevenly distributed across parameter groups.

The computational results also show that this adaptivity has non-negligible cost. In the 2000-iteration runs, the online and groupwise online samplers require substantially more wall-clock time than uniform sampling and static TASS. This is expected from the implementation: online methods repeatedly update discounted ridge regressions, compute feature-dependent proposal probabilities, and evaluate feedback responses from sampled block gradients. The groupwise variants further maintain separate online regressions for different parameter groups. We therefore interpret these methods as variance- and allocation-oriented procedures rather than as runtime improvements over static TASS.

6.2 Practical-proxy TASS RMSE sweep

Figure 3 plots the gradient RMSE and group-oracle ratio for each of the seven samplers across the seven perturbation magnitudes δ , and Table 3 gives the full numerical pivot. The vertical axis on the right panel is on a logarithmic scale because one of the static-method cells exceeds the others by an order of magnitude.

One notable thing is the static group-wise proxy is not monotonically bad as δ grows. Its oracle

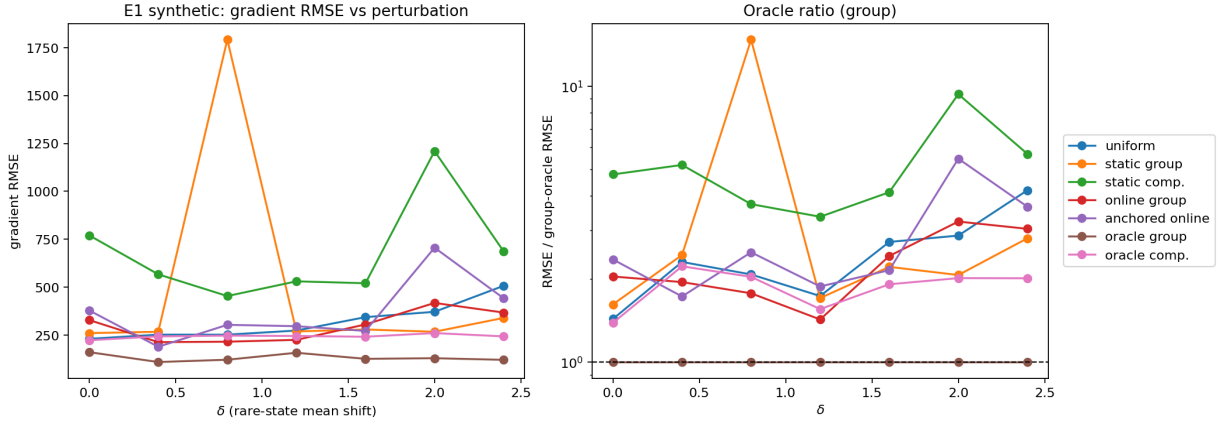


Figure 3: Practical-proxy TASS sweep on the stationary three-state synthetic HMM. Left: gradient RMSE versus the rare-state mean perturbation δ for each sampler. Right: oracle ratio against the group-wise oracle, on a log scale. The static group-wise proxy is competitive at most δ but blows up at $\delta = 0.8$; the online group-specific learner has no such spike.

δ	Uniform	Static gp.	Static comp.	Online gp.	Anchored online	Oracle gp.	Oracle comp.
0.0	1.43	1.62	4.79	2.04	2.35	1.00	1.39
0.4	2.31	2.44	5.18	1.95	1.73	1.00	2.22
0.8	2.08	14.76	3.74	1.78	2.50	1.00	2.04
1.2	1.73	1.71	3.36	1.43	1.88	1.00	1.56
1.6	2.73	2.21	4.13	2.42	2.15	1.00	1.91
2.0	2.87	2.07	9.36	3.23	5.46	1.00	2.01
2.4	4.20	2.81	5.68	3.04	3.66	1.00	2.01

Table 3: Oracle ratio $\text{RMSE}/\text{RMSE}_{\text{oracle group}}$ for the practical-proxy sweep on the synthetic three-state Gaussian HMM, averaged over $n_{\text{rep}} = 20$ replicates per cell. Bold marks the best non-oracle method at each δ . “Anchored online” is the proxy-anchored variant of online group-specific TASS; “Oracle comp.” is the component-wise oracle.

ratio is a reasonable 1.62 at the pilot, jumps to a catastrophic 14.76 at $\delta = 0.8$, recovers to 1.71 at $\delta = 1.2$, and then drifts slowly back up to 2.81 at $\delta = 2.4$. This is the empirical signature of a localized misalignment between the pilot’s K-means proxy labels and the current oracle’s importance distribution: the pilot happens to be most misaligned at one particular intermediate perturbation, and the static weights are pinned on the wrong side of the boundary for that slice of the sweep. Second, the static component-wise proxy is uniformly worse than the group-wise version, with oracle ratios between 3.36 and 9.36 and a second spike at $\delta = 2.0$. This is consistent with the prediction that splitting a fixed sampling budget across more coordinates pays a variance cost when the per-coordinate proxy weights are imprecise; the K-means proxy on this synthetic HMM is not precise enough to compensate for the smaller per-coordinate budget.

Online group-specific TASS occupies a different regime. Its oracle ratio is bounded between 1.43 and 3.23 across the entire sweep, with no analogue of the static group-wise spike. At every δ between 0.4 and 1.2 inclusive it is the best non-oracle method, and at $\delta = 0.8$ it converts what is the worst possible perturbation for the static method (14.76) into the second-best non-oracle ratio in the entire table (1.78). At the largest perturbations $\delta \in \{1.6, 2.0, 2.4\}$ the static group-wise method partially recovers and edges out online by 0.2–0.6 ratio points, but online remains the second-best

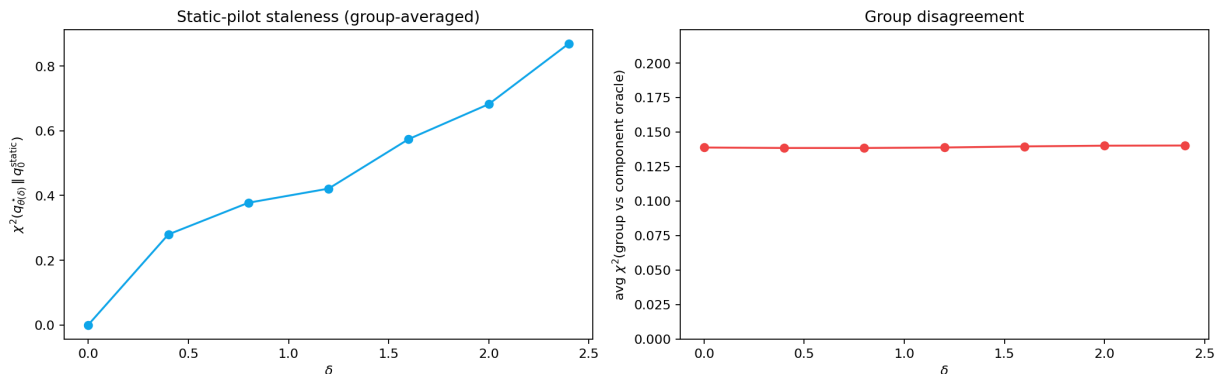


Figure 4: Mechanism-level diagnostics for the practical-proxy sweep. Left: group-averaged chi-square staleness of the static pilot weights versus the current oracle, monotonically growing from 0 to 0.87 across the sweep. Right: group disagreement, the average chi-square distance between the group-wise and component-wise oracle weights, which is essentially constant (0.138–0.140). The right panel says the strictness condition of Theorem 4.7 is satisfied uniformly on this experiment.

method in those cells and substantially outperforms both the static component-wise variant and the proxy-anchored online variant. At $\delta = 0$ uniform sampling is the best non-oracle method (1.43 versus online’s 2.04), repeating the negative-result-at-the-pilot phenomenon that also appears in the real-data experiment of Section 6.3: a frozen pilot is exactly as good as the proxy that built it.

The proxy-anchored variant of online group-specific TASS is unstable on this experiment in a way that mirrors the real-data finding. It improves on plain online at $\delta = 0.4$ (ratio 1.73 versus 1.95) and at $\delta = 1.6$ (ratio 2.15 versus 2.42), but it spikes to 5.46 at $\delta = 2.0$ and to 3.66 at $\delta = 2.4$. The instability comes from the multiplicative coupling between the static proxy and the online residual: when the proxy is locally well-aligned the anchor accelerates convergence, but when the proxy is locally misaligned the anchor amplifies that misalignment and the resulting low-probability blocks blow up the inverse-probability variance.

The group-averaged chi-square staleness $\overline{D}^{\text{static}}(\delta)$ grows monotonically from 0.00 at the pilot to 0.28, 0.38, 0.42, 0.57, 0.68, and 0.87 across the seven δ values, exactly as Corollary 4.4 predicts. The absolute scale here is two orders of magnitude smaller than the analogous quantity on the real BTC data (Section 6.3, 0 to 12.57), reflecting that this stationary synthetic HMM is sensitive to relatively small parameter shifts: the static RMSE penalty in this experiment is being driven less by chi-square distance and more by the absolute scale of the gradient at $\theta(\delta)$, which the static method does not track. The group disagreement, the average chi-square distance between the group-wise and component-wise oracle weights at $\theta(\delta)$, is essentially constant across the sweep at 0.138–0.140. This indicates that the gap between group-wise and component-wise oracle targets is a structural feature of the data-generating process rather than something that emerges as θ moves: the strictness condition in Theorem 4.7 is satisfied uniformly along the sweep, and any group-specific advantage we observe in the RMSE table is consistent with that theoretical prediction.

The group-versus-component oracle gap is itself the cleanest empirical illustration of Theorem 4.7 we have. Under the group-RMSE objective, the group oracle has lower RMSE than the component oracle by a factor of approximately 1.4 to 2.2 at every δ , so the component-wise scheme genuinely cannot match the group-aware target on this data-generating process. Online group-specific TASS sits between the two oracles in most cells, confirming that the online learner is approximating the group-aware target rather than collapsing onto the component-aware one. The

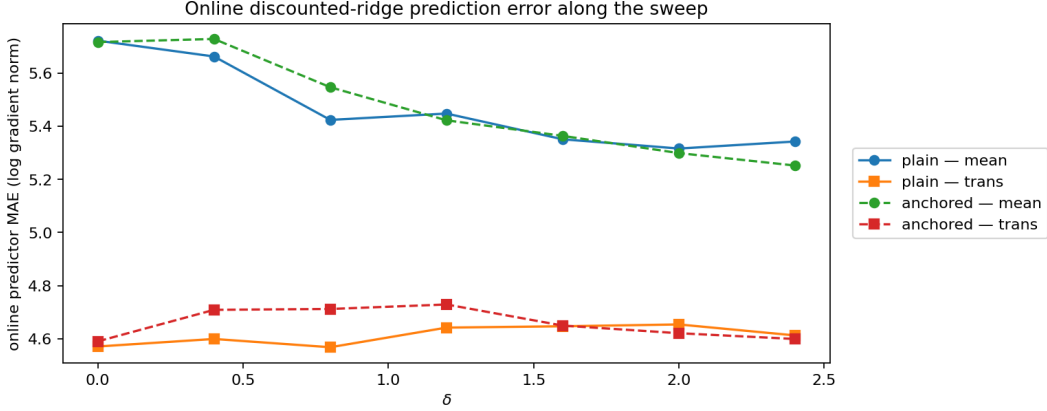


Figure 5: Mean absolute error of the online discounted-ridge predictor on the log block-gradient norm, broken out by parameter group (mean coordinates versus transition coordinates) and by variant (plain versus proxy-anchored). The predictor is far from precise in absolute terms but stays well-behaved across the sweep, which is all the near-oracle guarantee of Theorem 4.5 requires.

static group-wise method also lies in this band when its pilot labels happen to be well-aligned; what online buys us is the elimination of the bad-alignment cells, not a uniform improvement.

Finally, Figure 5 reports the online discounted-ridge predictor’s mean absolute error in the log block-gradient-norm prediction task that drives its sampling probabilities. The mean-coordinate MAE varies between 5.3 and 5.7 across the sweep and the transition-coordinate MAE between 4.5 and 4.7; the anchored and plain variants are within 0.2 MAE of each other in almost every cell. This is large in absolute terms but it is exactly what the near-oracle guarantee of Theorem 4.5 can tolerate, because the static method is paying the chi-square staleness penalty above and the online method is paying a log-MAE penalty that grows much more slowly with the perturbation. We guess this suggests that the online learner does not need to predict log block-gradient norms accurately; it only needs to predict them better than the static pilot’s chi-square error, and on this experiment it does.

6.3 Real-data results

We instantiate the protocol of Section 5.3 on $T = 1500$ bars (about five trading days) of 5-minute BTC USDT futures log realized variance, partitioned into $N \approx 70$ buffered blocks. At each $\delta \in \{0, 0.5, 1.0, 1.5, 2.0\}$ along the rare-state mean direction $v = e_{\mu_{\text{stress}}}$, both online learners are advanced six adapt steps from the state inherited at the previous δ , and then all six samplers are evaluated against the exact full block-gradient bundle over $n_{\text{rep}} = 50$ replicates. Figure 6 shows gradient RMSE and the group-oracle ratio as a function of δ , Table 4 reports the full pivot, and Figures 7 and 8 report the two diagnostic plots.

Online group-specific TASS is the best non-oracle method at every $\delta > 0$, and it is essentially tied with uniform at the pilot $\delta = 0$. Its oracle ratio is bounded between 1.30 and 1.69 across the entire $\delta > 0$ portion of the grid, while uniform is bounded between 1.64 and 2.03 on the same grid, so online cuts the variance gap to the oracle by a substantial margin at every shifted parameter value. The pattern is consistent with the near-oracle guarantee of Theorem 4.5 whenever the discounted-ridge predictor remains reasonably accurate, and is also consistent with what the analogous synthetic experiment of Section 6.2 found: the online learner is not always strictly dominant at the pilot, but it dominates at every cell where the static methods have to extrapolate.

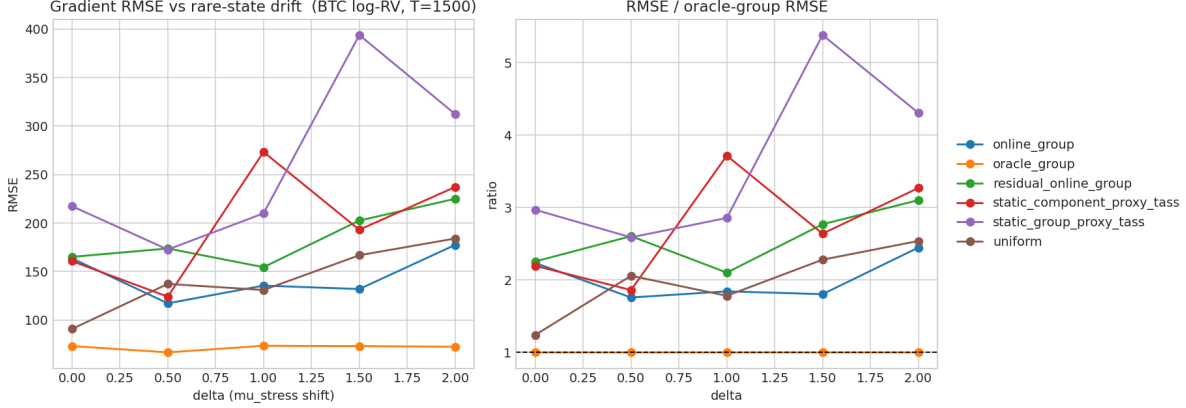


Figure 6: Real-data sweep on five trading days of 5-minute BTC log realized variance. Left: gradient RMSE versus the rare-state mean perturbation δ . Right: oracle ratio against the group-wise oracle. Online group-specific TASS is the best non-oracle method at every $\delta > 0$; the static group-wise and component-wise proxies degrade catastrophically at $\delta = 1.5$.

δ	Uniform	Static group	Static comp.	Online group	Anchored online	Oracle
0.0	1.50	2.14	2.00	1.55	2.13	1.00
0.5	1.92	2.67	2.48	1.47	2.33	1.00
1.0	1.74	2.72	5.52	1.30	2.48	1.00
1.5	2.03	17.15	14.92	1.69	2.61	1.00
2.0	1.64	2.31	3.13	1.56	3.28	1.00

Table 4: Oracle ratio $\text{RMSE}/\text{RMSE}_{\text{oracle}}$ for each sampler on the real BTC log-RV sweep, averaged over $n_{\text{rep}} = 50$ replicates per cell using common random numbers across methods so the comparison variance is reduced. Bold marks the best non-oracle method at each δ . “Static group” and “Static comp.” are the frozen proxy-TASS pilot weights of Section 3.5; “Anchored online” is the proxy-anchored variant of online group-specific TASS.

The big result is the catastrophic blowup of both static methods at $\delta = 1.5$. The static group-wise proxy reaches oracle ratio 17.15 at $\delta = 1.5$, and the static component-wise proxy reaches 14.92; this is roughly a tenfold gap relative to online group-specific TASS, which sits at 1.69 in the same cell. The intermediate $\delta = 1.0$ also shows a smaller version of the same effect, with static component-wise at 5.52 versus online at 1.30. These cells correspond to the parameter values at which the K-means proxy labels used to construct the static pilot weights are most mismatched with the current oracle’s importance distribution, and they are precisely where Corollary 4.4 predicts the static estimator’s variance penalty should be largest. Online group-specific TASS converts the worst-case static cell into the cleanest non-oracle cell in the table, which is the strongest single piece of evidence in the paper that adapting block weights from gradient feedback pays off when it most matters.

Figure 7 reports the group-averaged chi-square staleness $\bar{D}^{\text{static}}(\delta)$ of the static pilot weights against the current oracle. It rises monotonically from 0.00 at $\delta = 0$ to 7.65, 10.77, 11.67, and 12.57 across the perturbation grid, exactly as Corollary 4.4 predicts. The monotone growth of the staleness in δ tracks the monotone degradation of the static methods in Table 4 up to $\delta = 1.5$. The partial recovery of the static group-wise method at $\delta = 2.0$ (ratio 2.31, well below the 17.15 at $\delta = 1.5$) shows that staleness alone does not determine the variance penalty: the absolute

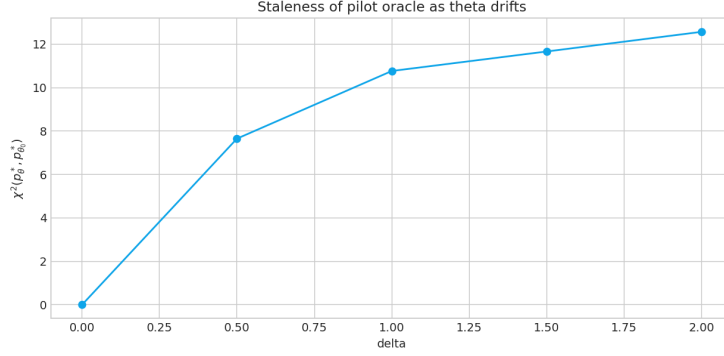


Figure 7: Group-averaged chi-square staleness of the proxy-TASS pilot weights relative to the oracle weights at $\theta(\delta)$. Monotone growth from 0 to 12.6 along the sweep is exactly the quantity Corollary 4.4 predicts will appear in the static-TASS variance penalty.

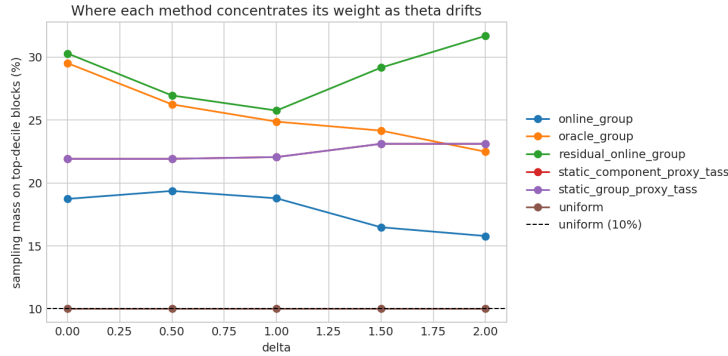


Figure 8: Top-decile sampling mass, $\sum_{n \in \text{top}_{10\%}} p_n$, with the top decile chosen by the true block-gradient norm at $\theta(\delta)$. Uniform stays at the floor 0.10 by construction. The anchored online variant over-concentrates relative to the oracle, which Table 4 shows is paid for in $1/p_n$ variance.

scale of the gradient at $\theta(\delta)$ also matters, and at $\delta = 2.0$ the perturbed configuration happens to be coincidentally easier for the static method’s pilot to track than the worst-case $\delta = 1.5$ configuration.

The proxy-anchored variant of online group-specific TASS is consistently worse than the unanchored version, with oracle ratios 2.13, 2.33, 2.48, 2.61, and 3.28 versus the unanchored 1.55, 1.47, 1.30, 1.69, 1.56. Figure 8 shows the underlying mechanism. The anchored sampler places more probability mass on the true top-decile gradient-norm blocks than even the oracle does (0.27–0.32 versus the oracle’s 0.22–0.30), while the unanchored online learner places less (0.16–0.19). The anchored sampler is therefore over-concentrated, and the resulting low-probability blocks blow up the inverse-probability variance even though the targeting story is favorable. This is a useful empirical reminder that top-decile mass is necessary but not sufficient for low gradient RMSE: the variance penalty in Section 4.2 weights $1/p_n$ as well as $\|g_n\|^2$, and the multiplicative anchor concentrates mass without controlling the resulting tail of $1/p_n$.

A small but noteworthy result is the comparison at the pilot $\delta = 0$. Uniform sampling has oracle ratio 1.50, online group-specific TASS has 1.55, and every static-TASS variant sits at 2.00 or higher. The static pilot is built from a single-pass posterior-state proxy rather than from oracle block gradients, so at the pilot itself its concentration is targeted at the wrong notion of importance, and uniform sampling slightly wins because no method has had time to learn a better target. By

$\delta = 0.5$ the online learners have absorbed enough feedback to overtake uniform, while the static pilots inherit the misalignment permanently. The frozen pilot weights are exactly as good as the proxy that built them and no better, in agreement with the analogous result in Section 6.2.

The methodological claim of the paper does not require that the rare state be high-volatility or rare in any strict economic sense, only that some blocks carry disproportionate gradient information for some parameter groups and that this set drifts with θ . Both conditions are clearly met on this data: the staleness curve grows by more than an order of magnitude across the sweep, and online group-specific TASS converts that drift into a roughly tenfold RMSE reduction over the static baseline at the worst-case cell.

7 Discussion

This project started from the observation from TASS, where in HMMs with rare or high-information regimes, not all blocks are equally useful for estimating every parameter. Static TASS already takes this seriously by assigning larger sampling mass to blocks that appear informative for particular latent-state components. Our results confirm that this is a strong idea. In the one-rare-state changepoint experiment, TASS remains the best or nearly best method across most shifted regimes, even though the data-generating mean changes halfway through the sequence. In the practical-proxy synthetic sweep, the static group-wise proxy is also competitive in many cells. So TASS is a strong baseline precisely because a good proxy for rare-state information can go a long way.

The reason to introduce online and group-specific adaptation is that the proxy is not always good in the same way throughout the run. In the changepoint experiment with two rare states, static TASS often appears to favour one rare component and do badly on the other. In the practical-proxy sweep, the static group-wise method has a sharp failure at an intermediate perturbation, while the online group-specific method avoids that spike. Also in the BTC log-realised-variance experiment, once the stress-state mean is perturbed away from the pilot value, online group-specific TASS is the best non-oracle method at every shifted value of δ , while the static proxy methods can deteriorate very badly. So overall empirics suggest that online TASS does not replace static TASS in every regime, but it does give a way to recover when the fixed pilot weights stop matching the blocks that currently carry the gradient information (which is what we hoped for, even if our way to get to this was roundabout).

There are several things we would want to improve before treating this as a general-purpose sampler. A big remaining point was that our initial intention was to work on the full posterior, but after wrangling with this for many days the problem seems hard to track with. Perhaps a stab with a different set of financial data more conducive to the setting (the stuck pipe scenario is interesting, as a latent variable for local pricing, perhaps?) would be tractable. Second, our online regression model is pretty simple, where we just discounted ridge on hand-built block features. Were the underlying theory to be richer, I suppose we could be less ad hoc with e.g. parameter grouping, or when/how/strong we want our online update to be. In this sense developing theory (and e.g. analyzing how we want to spend compute and wall-clock time, perhaps thinking about computational cost theoretically) could be useful. Also, perhaps BTC testing should be expanded beyond a single BTC window and a one-dimensional perturbation of the stress-state mean, so testing on longer market sequences, clinical time series, or industrial fault data will help make relevant blocks' information change organically rather than through a perturbation.

A Implementation details

A.1 Our compared (sub)samplers

All samplers were implemented on the same blocked representation of the observation sequence. For a fixed radius L , the training sequence was divided into consecutive blocks of length $2L + 1$. In the experiments reported here $L = 2$, so each block contained five observations. If N denotes the resulting number of complete blocks, then all proposal distributions used by the samplers were probability vectors on $\{1, \dots, N\}$. Incomplete terminal observations, if present, were discarded before constructing the blockwise proposal distributions.

At each SGLD iteration we updated the transition parameters and the emission parameters separately. The transition update used a minibatch of blocks to estimate the gradient contribution for the transition matrix. The emission update used minibatches of blocks to estimate the gradient contributions for the Gaussian means and variances. For each sampled block n , the implementation computed the blockwise gradients

$$g_n(\theta_t) = (g_{n,\mu}(\theta_t), g_{n,\sigma^2}(\theta_t), g_{n,A}(\theta_t)) ,$$

using the local forward-backward approximation with buffer parameter B . In the reported runs $B = 5$. When the same block was requested more than once inside a single gradient-estimation call, its block-gradient tuple was cached and reused, so that repeated samples did not trigger repeated local likelihood calculations.

For all non-uniform proposal distributions, the sampled block contribution was divided by the probability with which that block was sampled. Thus, if a scalar or groupwise proposal distribution assigned probability p_n to block n , the corresponding block contribution was multiplied by $1/\max(p_n, 10^{-300})$ before being accumulated. This lower numerical bound was used only to avoid division by zero in floating-point arithmetic. After summing the minibatch contributions, the estimate was divided by the minibatch size. In the experiments reported here the minibatch size was

$$m_{\text{batch}} = 5.$$

The prior-gradient contribution was added before the blockwise data contributions were accumulated.

The **uniform** sampler used the same probability vector for every parameter component:

$$p_n = \frac{1}{N}, \quad n = 1, \dots, N.$$

For compatibility with the other implementations, the uniform probabilities were stored separately for the mean coordinates, variance coordinates, and transition coordinates, but all entries were identical.

The **tass** sampler used a static blockwise proposal computed once before the chain was run. The implementation first applied K -means clustering to the training observations, with $K = 3$. The resulting proxy labels were used to construct blockwise counts, blockwise empirical means, blockwise within-cluster variance summaries, and blockwise transition counts. For each mean coordinate j , the unnormalised score of block n was

$$c_{n,j} |\bar{y}_{n,j} - \bar{y}_j| + \eta^2,$$

where $c_{n,j}$ is the number of observations in block n assigned to cluster j , $\bar{y}_{n,j}$ is the within-block cluster mean, and $\bar{y}_j$ is the corresponding full-training proxy mean. For each variance coordinate j , the unnormalised score was

$$c_{n,j} (s_j^2 + s_{n,j}^2) + \eta^2,$$

where s_j^2 is the global proxy variance and $s_{n,j}^2$ is the corresponding blockwise within-cluster term. For each transition coordinate (j, k) , the unnormalised score was computed from the number of proxy transitions from k to j in the block, divided by the corresponding smoothed proxy transition probability. All static scores were normalised over blocks before use. In the reported experiments $\eta = 10^{-5}$.

The **online** sampler used a single online discounted ridge model to define one global block proposal. Each block n was assigned a feature vector w_n . The implemented feature vector contained an intercept, a normalised block-time coordinate, the block-level changepoint indicator, blockwise empirical moments and tail summaries of the observations, and interaction terms between the time/changepoint coordinates and selected blockwise moment features. All non-intercept features were standardised across blocks before fitting. At iteration t , the fitted coefficient vector $\hat{\beta}_t$ induced the proposal

$$p_{n,t} = (1 - \lambda) \frac{\exp\{\frac{1}{2} w_n^\top \hat{\beta}_t\}}{\sum_{\ell=1}^N \exp\{\frac{1}{2} w_\ell^\top \hat{\beta}_t\}} + \frac{\lambda}{N}.$$

In the main runs,

$$\rho = 0.95, \quad \tau = 10, \quad \lambda = 0.10.$$

The linear scores $w_n^\top \hat{\beta}_t$ were clipped to the interval $[-30, 30]$ before exponentiation. After each sampled block was evaluated, the online response was formed as

$$f_{t,n} = \log \left(\|g_n(\theta_t)\|^2 + \epsilon_{\text{fb}} \right),$$

where the norm includes the mean, variance, and transition block-gradient components. The discounted ridge sufficient statistics were then updated by

$$G_t = \rho G_{t-1} + w_n w_n^\top, \quad h_t = \rho h_{t-1} + w_n f_{t,n},$$

and the coefficient vector was recomputed from the ridge system

$$\hat{\beta}_t = (G_t + \tau I)^{-1} h_t.$$

In the implementation the system was solved by a jittered linear solve, with a pseudoinverse fallback if the solve failed.

The **online_tass** sampler used the same global online update as **online**. The only implementation difference was the initial value of the ridge model. Before the chain began, a static TASS proxy score was computed for each block using the same K -means proxy construction as in **tass**. These static scores were converted into pseudo-responses

$$\tilde{f}_n^{\text{static}} = \log \left((\tilde{a}_n^{\text{static}})^2 + \epsilon_{\text{fb}} \right),$$

and the initial ridge coefficient was obtained by fitting the ridge system to all block features and pseudo-responses. After this initialisation step, the sampler used the same online probability update and the same online feedback rule as **online**. The static TASS scores were not multiplied into the proposal after initialisation.

The **online_group** sampler used three separate online discounted ridge models, one for each parameter group

$$r \in \{\mu, \sigma^2, A\}.$$

Each group had its own coefficient vector $\hat{\beta}_t^{(r)}$, sufficient statistics, and probability vector

$$p_{n,t}^{(r)} = (1 - \lambda) \frac{\exp\{\frac{1}{2} w_n^\top \hat{\beta}_t^{(r)}\}}{\sum_{\ell=1}^N \exp\{\frac{1}{2} w_\ell^\top \hat{\beta}_t^{(r)}\}} + \frac{\lambda}{N}.$$

The transition update sampled blocks from $p_t^{(A)}$. The mean update sampled blocks from $p_t^{(\mu)}$. The variance update sampled blocks from $p_t^{(\sigma^2)}$. The response used to update the group r was

$$f_{t,n}^{(r)} = \log (\|P_r g_n(\theta_t)\|^2 + \epsilon_{\text{fb}}) ,$$

where P_r denotes selection of the corresponding block-gradient group. Within a group, the same block probability vector was used for all scalar coordinates in that group.

The **online_tass_group** sampler used the same groupwise online implementation as **online_group**. Its only difference was the initialisation of the three ridge models. For each group $r \in \{\mu, \sigma^2, A\}$, the implementation computed static group scores

$$\tilde{a}_{n,r}^{\text{static}}$$

from the K -means proxy labels and the initial pilot parameter value. These scores were converted to pseudo-responses

$$\tilde{f}_{n,r}^{\text{static}} = \log ((\tilde{a}_{n,r}^{\text{static}})^2 + \epsilon_{\text{fb}}) ,$$

and a separate ridge coefficient $\hat{\beta}_0^{(r)}$ was fitted for each group. Subsequent sampling probabilities were then determined entirely by the online groupwise ridge updates. As in the global TASS-initialised sampler, the static proxy affected only the initial ridge state.

All online samplers used an optional initial warmup period during which uniform block probabilities were used. In the main configuration this warmup length was set to 100 iterations. During the chain, the transition parameter was represented through an unconstrained positive matrix $\hat{A}$, and the stochastic update was followed by column normalisation to obtain a valid transition matrix A . This normalisation was also repeated every ten iterations. Variance updates that produced non-positive or non-finite variance values were rejected by restoring the previous variance vector. The full chain stored μ_t , σ_t^2 , A_t , $\hat{A}_t$, the sampler state, and the wall-clock runtime for each method.

A.2 Changepoint staleness synthetic task

The main experiments use $T_{\text{train}} = 50,000$ training observations and $T_{\text{test}} = 50,000$ held-out observations. With $L = 2$, this gives 10,000 training blocks and 10,000 held-out blocks. We take $K = 3$, block size $2L + 1 = 5$, minibatch size 5, SGLD step size 10^{-5} , and 2000 SGLD iterations. The default online ridge settings in the main run are

$$\rho = 0.95, \quad \tau = 10, \quad \lambda = 0.10,$$

where ρ is the discount factor, τ is the ridge penalty, and λ is the uniform exploration floor in the online sampling distribution. For the online methods, the first warmup iterations use uniform sampling before ridge feedback is used. Linear systems in the discounted ridge updates are solved with a small diagonal jitter.

The online regression features w_n are block-level summaries. They include an intercept, a normalised block index, the post-changepoint block indicator, the block mean, mean absolute value, standard deviation, mean square, maximum absolute value, a tail-count feature, the fraction of negative observations, and interactions between time or changepoint information and magnitude features. All non-intercept features are standardised over the training blocks.

Because the hidden-state labels are exchangeable, diagnostics are computed after sorting sampled emission means. For a rare state k , we determine its rank in the sorted vector of effective

training means and compare it with the corresponding sorted coordinate of the sampled emission means. If μ_k^{eff} denotes the effective training mean of state k , the final mean error is

$$\left| \mu_{T,r(k)}^{\text{sorted}} - \mu_k^{\text{eff}} \right|,$$

where $r(k)$ is the rank of state k among the effective means. We also report the average of this error over the second half of the chain.

For each rare state, we evaluate held-out predictive performance on observations whose latent state is known from the simulation. Given posterior samples up to iteration c , the predictive density assigned to a held-out observation y from state k is estimated by

$$\hat{p}_c(y | k) = \frac{1}{c+1} \sum_{t=0}^c \varphi\left(y; \mu_{t,r(k)}^{\text{sorted}}, (\sigma^2)_{t,r(k)}^{\text{sorted}}\right),$$

where $\varphi(\cdot; m, s^2)$ is the univariate Gaussian density. We compute this quantity in log space using the log-mean-exp identity. The reported log-predictive metric is the average of $\log \hat{p}_c(y | k)$ over a held-out subset of observations from the target state.

We report three held-out variants. The “all” metric averages over held-out observations from the rare state. The “high-information” metric restricts to post-changepoint rare-state observations. The “low-information” metric restricts to pre-changepoint rare-state observations. In the main changepoint experiments, the high-information metric is the main held-out diagnostic because the intervention is precisely a post-changepoint change in the rare-state mean.

To assess the quality of a block sampling distribution independently of the downstream posterior chain, we compute a second-moment diagnostic for the rare-state mean gradient. Let s_n denote the block-gradient contribution for the relevant rare-state mean at the final iterate. For a sampling distribution p over blocks, define

$$Q(p) = \sum_{n=1}^N \frac{s_n^2}{p_n}.$$

The oracle distribution for this one-dimensional diagnostic is

$$p_n^* = \frac{|s_n|}{\sum_{j=1}^N |s_j|},$$

and the reported diagnostic is

$$\frac{Q(p)}{Q(p^*)}.$$

Values close to one indicate that the method assigns mass in approximate proportion to the relevant gradient magnitudes. Large values indicate that the method undersamples blocks that contribute substantially to the corresponding gradient component.

We also report the sampling mass assigned to post-changepoint rare blocks. Let p_n be the block probability used by a method for the relevant rare-state mean component, and let $c_{n,k}$ be the number of observations in block n with latent state k . Let $h_n \in [0, 1]$ be the block-average post-changepoint indicator. For a rare state k , the high-information rare-block mass is

$$\sum_{n=1}^N p_n \mathbf{1}\{c_{n,k} > 0\} \mathbf{1}\{h_n \geq 0.5\}.$$

This quantity records whether a method is placing sampling mass on blocks that both contain the rare state and occur after the changepoint.

Wall-clock time is recorded for each method in each scenario.

A.3 Practical-proxy TASS RMSE sweep

We generate $T = 2400$ observations from a stationary three-state multivariate Gaussian HMM with observation dimension $D = 3$ and one rare state, produced by the helper `make_three_state_rare_gaussian_hmm`. The covariance matrices Σ_k are treated as known and held fixed throughout the experiment. The unknown parameter is the concatenation of the emission means $\mu \in \mathbb{R}^{K \times D}$ and the transition logits $\beta_A \in \mathbb{R}^{K \times K}$, with $K = 3$. The pilot θ_0 is set to the true data-generating parameter, and the sweep direction v_{rare} is the unit vector that increases the rare-state mean along its first observation coordinate.

The training sequence is partitioned into non-overlapping central blocks of half-width $h = 6$, so each block has length $2h + 1 = 13$. Each block is padded with a buffer of $B = 4$ observations on either side for the local forward-backward recursion. The resulting number of complete blocks is $N \approx 184$. Block features $w_n \in \mathbb{R}^{16}$ are produced by `compute_block_features` and include, for each block, the per-coordinate empirical mean and variance of the observations, the per-coordinate minimum and maximum, the ℓ_2 norm of the block-mean vector, and a small number of cross-coordinate interaction terms. All non-constant features are mean-standard-deviation normalized across blocks before being passed to the online ridge predictors. The minibatch size used by every sampler is

$$M_{\text{mean}} = M_{\text{trans}} = 6$$

blocks per parameter group per gradient-estimation call. The static component-wise proxy uses the derived budget

$$M_{r,d} = \max\{1, \lfloor KM_r/(KD) \rfloor\} = 2$$

blocks per component, so its total per-step gradient evaluations match the group-wise methods.

The sweep grid is $\delta \in \{0.0, 0.4, 0.8, 1.2, 1.6, 2.0, 2.4\}$ and is processed in order. For each δ , the experiment proceeds in three stages. First, both the plain and proxy-anchored online ridge models run $S = 10$ adapt steps of the online group-specific TASS update, starting from the model state inherited from the previous δ . Each adapt step draws M_{mean} blocks from the current per-group sampling distribution, evaluates the buffered block gradients at $\theta(\delta)$, updates the discounted-ridge sufficient statistics on the response

$$f_{t,n}^{(r)} = \log(\|P_r g_n(\theta(\delta))\|_2^2 + \epsilon),$$

and refits the ridge coefficient $\hat{\beta}^{(r)}$ for that group. Second, gradient RMSE is evaluated for each sampler over $n_{\text{rep}} = 20$ replicates, using the same set of buffered block gradients across all samplers and across all replicates within a sampler so that variance reduction across the comparison is maximal. Third, the chi-square staleness of the static group-wise pilot weights against the current oracle is computed exactly, along with the average chi-square distance between the group-wise and component-wise oracle weights at $\theta(\delta)$.

The online ridge predictor has discount $\rho = 0.99$, ridge penalty $\tau = 50$, uniform exploration floor $\lambda = 0.05$ (so the final sampling distribution is $(1 - \lambda)\tilde{p}_n + \lambda/N$), and log-feedback floor $\varepsilon = 10^{-8}$. Linear scores $w_n^\top \hat{\beta}^{(r)}$ are clipped to a symmetric range before exponentiation, and the ridge system is solved with a small diagonal jitter so the solver remains stable even when a block is sampled repeatedly within a single adapt step. The proxy-anchored variant differs from the plain variant only in that the per-group sampling distribution is rescaled by the pilot proxy weights raised to the power $\eta = 1.0$ before normalization. The static group-wise and component-wise proxy weights are constructed once, before the sweep begins, using the same K-means proxy construction described in Appendix A.2 ($K_{\text{km}} = 3$, $\eta_{\text{score}} = 10^{-5}$) and mixed with a 10^{-3} uniform component for numerical safety.

The group-averaged chi-square staleness reported in Figure 4 is

$$\overline{D}^{\text{static}}(\delta) = \frac{1}{R} \sum_{r=1}^R \chi^2(q_{r,\theta(\delta)}^* \parallel q_{0,r}^{\text{static}}),$$

averaged over the parameter groups $r \in \{\mu, \beta_A\}$. The group disagreement is

$$\overline{D}^{\text{group}}(\delta) = \frac{1}{RKD} \sum_{r,k,d} \chi^2(q_{r,k,d,\theta(\delta)}^* \parallel q_{r,\theta(\delta)}^*),$$

the mean chi-square distance between each component's oracle weights and the group-wise oracle to which that component belongs. The two diagnostics share no Monte Carlo noise with the RMSE estimates because they are functions of the deterministic oracle weight bundles at $\theta(\delta)$.

B Proofs

B.1 Proof of Proposition 4.1

For the one-step statement, couple the full-gradient and stochastic-gradient transitions using the same Gaussian noise ξ_t . Conditional on $\mathcal{F}_t$, define

$$\bar{\theta}_{t+1} = \theta_t + \frac{\eta_t}{2} G(\theta_t) + \sqrt{\eta_t} \xi_t,$$

and

$$\theta_{t+1} = \theta_t + \frac{\eta_t}{2} \widehat{G}_t(\theta_t) + \sqrt{\eta_t} \xi_t.$$

Then

$$\theta_{t+1} - \bar{\theta}_{t+1} = \frac{\eta_t}{2} \{\widehat{G}_t(\theta_t) - G(\theta_t)\}.$$

Taking conditional second moments gives

$$\mathbb{E}[\|\theta_{t+1} - \bar{\theta}_{t+1}\|_2^2 \mid \mathcal{F}_t] \leq \frac{\eta_t^2}{4} \sigma_t^2.$$

The squared Wasserstein distance is bounded by the squared distance under any coupling, so the same upper bound applies to

$$W_2^2(K_t^{\text{sg}}(\theta_t, \cdot), K_t^{\text{full}}(\theta_t, \cdot)).$$

For the finite-horizon statement, use the coupled recursions

$$\theta_{t+1} = \theta_t + \frac{\eta_t}{2} \widehat{G}_t(\theta_t) + \sqrt{\eta_t} \xi_t, \quad \bar{\theta}_{t+1} = \bar{\theta}_t + \frac{\eta_t}{2} G(\bar{\theta}_t) + \sqrt{\eta_t} \xi_t.$$

Let $\Delta_t = \theta_t - \bar{\theta}_t$ and

$$\zeta_t = \widehat{G}_t(\theta_t) - G(\theta_t).$$

Then

$$\Delta_{t+1} = \Delta_t + \frac{\eta_t}{2} \{G(\theta_t) - G(\bar{\theta}_t)\} + \frac{\eta_t}{2} \zeta_t.$$

Conditional on $\mathcal{F}_t$, the first two terms are deterministic and $\mathbb{E}[\zeta_t \mid \mathcal{F}_t] = 0$, so the cross term vanishes after conditioning. Therefore

$$\begin{aligned} \mathbb{E}[\|\Delta_{t+1}\|_2^2 \mid \mathcal{F}_t] &= \left\| \Delta_t + \frac{\eta_t}{2} \{G(\theta_t) - G(\bar{\theta}_t)\} \right\|_2^2 + \frac{\eta_t^2}{4} \mathbb{E}[\|\zeta_t\|_2^2 \mid \mathcal{F}_t] \\ &\leq \left(1 + \frac{L\eta_t}{2}\right)^2 \|\Delta_t\|_2^2 + \frac{\eta_t^2}{4} \sigma_t^2, \end{aligned}$$

where the Lipschitz bound gives

$$\|G(\theta_t) - G(\bar{\theta}_t)\|_2 \leq L \|\Delta_t\|_2.$$

Taking expectations and iterating the recursion from $\Delta_0 = 0$ yields

$$\mathbb{E} \|\Delta_T\|_2^2 \leq \sum_{s=0}^{T-1} \frac{\eta_s^2}{4} \mathbb{E} \sigma_s^2 \prod_{u=s+1}^{T-1} \left(1 + \frac{L\eta_u}{2}\right)^2.$$

The final simplified bound follows by replacing each η_t by η and each σ_t^2 by σ^2 .

B.2 Proof of Proposition 4.2

For a fixed group r , conditional on $\mathcal{F}_t$, the distribution $p_t^{(r)}$ is fixed. For one draw $I^{(r)} \sim p_t^{(r)}$,

$$\mathbb{E} \left[\frac{P_r g_{I^{(r)}}(\theta_t)}{p_{I^{(r)},t}^{(r)}} \mid \mathcal{F}_t \right] = \sum_{n=1}^N p_{n,t}^{(r)} \frac{P_r g_n(\theta_t)}{p_{n,t}^{(r)}} = \sum_{n=1}^N P_r g_n(\theta_t).$$

Averaging M_r independent draws does not change the conditional expectation, so

$$\mathbb{E}[\hat{G}_{t,r} \mid \mathcal{F}_t] = P_r \nabla \log \pi(\theta_t) + \sum_{n=1}^N P_r g_n(\theta_t).$$

Since the coordinate groups form a partition, $\sum_r P_r^\top P_r = I_d$. Therefore

$$\begin{aligned} \mathbb{E}[\hat{G}_t \mid \mathcal{F}_t] &= \sum_{r=1}^R P_r^\top \mathbb{E}[\hat{G}_{t,r} \mid \mathcal{F}_t] \\ &= \nabla \log \pi(\theta_t) + \sum_{n=1}^N g_n(\theta_t) = G(\theta_t). \end{aligned}$$

B.3 Proof of Theorem 4.3

Fix group r and suppress r in the notation. Write $a_n = a_{n,r}(\theta)$ and $A = \sum_n a_n$. For any distribution p ,

$$Q_\theta(p) = \sum_{n=1}^N \frac{a_n^2}{p_n} = A^2 \sum_{n=1}^N \frac{(a_n/A)^2}{p_n} = A^2 \sum_{n=1}^N \frac{(q_\theta^*(n))^2}{p_n}.$$

By definition,

$$\chi^2(q_\theta^* \| p) = \sum_{n=1}^N \frac{(q_\theta^*(n) - p_n)^2}{p_n} = \sum_{n=1}^N \frac{(q_\theta^*(n))^2}{p_n} - 1,$$

because $\sum_n q_\theta^*(n) = \sum_n p_n = 1$. Hence

$$Q_\theta(p) = A^2 \{1 + \chi^2(q_\theta^* \| p)\}.$$

The chi-square divergence is nonnegative and equals zero if and only if $p = q_\theta^*$, so q_θ^* is the unique minimizer of $Q_\theta(p)$ among positive distributions.

For the MSE identity, let $I \sim p$ and define

$$\hat{H}_p(\theta) = \frac{h_I(\theta)}{p_I}.$$

Then

$$\mathbb{E}\hat{H}_p(\theta) = \sum_{n=1}^N p_n \frac{h_n(\theta)}{p_n} = \sum_{n=1}^N h_n(\theta) = H(\theta).$$

Thus

$$\mathbb{E} \left\| \hat{H}_p(\theta) - H(\theta) \right\|_2^2 = \mathbb{E} \left\| \hat{H}_p(\theta) \right\|_2^2 - \|H(\theta)\|_2^2.$$

The first term is

$$\mathbb{E} \left\| \hat{H}_p(\theta) \right\|_2^2 = \sum_{n=1}^N p_n \frac{\|h_n(\theta)\|_2^2}{p_n^2} = \sum_{n=1}^N \frac{a_n^2}{p_n} = Q_\theta(p).$$

Averaging M_r independent copies divides the covariance, and hence the MSE, by M_r .

B.4 Proof of Corollary 4.4

Apply Theorem 4.3 with $p = q_{0,r}^{\text{static}}$ and $\theta = \theta_t$:

$$Q_{r,\theta_t}(q_{0,r}^{\text{static}}) = A_r(\theta_t)^2 \left\{ 1 + \chi^2(q_{r,\theta_t}^* \| q_{0,r}^{\text{static}}) \right\}.$$

The oracle value is $Q_{r,\theta_t}(q_{r,\theta_t}^*) = A_r(\theta_t)^2$. Substituting

$$D_{t,r}^{\text{static}} = \chi^2(q_{r,\theta_t}^* \| q_{0,r}^{\text{static}})$$

gives the claim.

B.5 Proof of Theorem 4.5

Fix group r and time t , and suppress both indices when convenient. Let

$$a_n = a_{n,r}(\theta_t), \quad b_n = \sqrt{a_n^2 + \varepsilon_r}, \quad B = \sum_{n=1}^N b_n.$$

The regularized oracle distribution is

$$q_n^\varepsilon = \frac{b_n}{B}.$$

The score accuracy assumption says

$$|\hat{m}_n - \log(a_n^2 + \varepsilon_r)| \leq \eta$$

for all n . Exponentiating half of this inequality gives

$$e^{-\eta/2} b_n \leq e^{\hat{m}_n/2} \leq e^{\eta/2} b_n.$$

After normalizing over n , the softmax distribution

$$\tilde{p}_n = \frac{e^{\hat{m}_n/2}}{\sum_j e^{\hat{m}_j/2}}$$

satisfies

$$e^{-\eta} q_n^\varepsilon \leq \tilde{p}_n \leq e^\eta q_n^\varepsilon \quad \text{for all } n.$$

The actual online distribution has a uniform floor,

$$p_n^{\text{online}} = (1 - \lambda) \tilde{p}_n + \lambda/N,$$

so $p_n^{\text{online}} \geq (1 - \lambda) \tilde{p}_n$. Therefore

$$\frac{1}{p_n^{\text{online}}} \leq \frac{1}{(1 - \lambda) \tilde{p}_n} \leq \frac{e^\eta}{(1 - \lambda) q_n^\varepsilon}.$$

Multiplying by a_n^2 and summing gives

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \frac{e^\eta}{1 - \lambda} Q_{r,\theta_t}(q^\varepsilon).$$

It remains to compare $Q(q^\varepsilon)$ with the true oracle value. Since

$$a_n \leq b_n \leq a_n + \sqrt{\varepsilon_r},$$

we have

$$A = \sum_n a_n \leq B \leq A + N\sqrt{\varepsilon_r}.$$

Moreover,

$$Q(q^\varepsilon) = \sum_n \frac{a_n^2}{b_n/B} = B \sum_n \frac{a_n^2}{b_n} \leq B \sum_n a_n = AB.$$

The true oracle value is $Q(q^*) = A^2$. Hence

$$Q(q^\varepsilon) \leq \frac{B}{A} Q(q^*) \leq \left(1 + \frac{N\sqrt{\varepsilon_r}}{A}\right) Q(q^*).$$

Combining the inequalities yields

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \frac{e^{\eta_{t,r}}}{1 - \lambda} \left(1 + \frac{N\sqrt{\varepsilon_r}}{A_r(\theta_t)}\right) Q_{r,\theta_t}(q_{r,\theta_t}^*).$$

B.6 Proof of Corollary 4.6

By Theorem 4.5,

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \alpha_{t,r} Q_{r,\theta_t}(q_{r,\theta_t}^*).$$

By Corollary 4.4,

$$Q_{r,\theta_t}(q_{0,r}^{\text{static}}) = (1 + D_{t,r}^{\text{static}}) Q_{r,\theta_t}(q_{r,\theta_t}^*).$$

Therefore, if $\alpha_{t,r} < 1 + D_{t,r}^{\text{static}}$, then

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) < Q_{r,\theta_t}(q_{0,r}^{\text{static}}).$$

The MSE comparison follows because both group estimators have the same subtractive constant $\|H_r(\theta_t)\|_2^2$ in the MSE formula.

B.7 Proof of Theorem 4.7

For group-specific sampling, each group r can choose its own oracle distribution

$$q_r^*(n) = \frac{a_{n,r}}{\sum_j a_{j,r}}.$$

Thus the optimal group-specific objective is

$$\mathcal{Q}_{\text{group}}^*(\theta) = \sum_r \omega_r \left(\sum_n a_{n,r} \right)^2.$$

If a single shared distribution p must be used for all groups, then the objective is

$$\mathcal{Q}_{\text{shared}}(p) = \sum_r \omega_r \sum_n \frac{a_{n,r}^2}{p_n} = \sum_n \frac{\sum_r \omega_r a_{n,r}^2}{p_n}.$$

By Theorem 4.3 applied to the scalar block magnitudes

$$c_n = \sqrt{\sum_r \omega_r a_{n,r}^2},$$

the shared oracle value is

$$\mathcal{Q}_{\text{shared}}^*(\theta) = \left(\sum_n c_n \right)^2 = \left(\sum_n \sqrt{\sum_r \omega_r a_{n,r}^2} \right)^2.$$

To compare the two values, define vectors

$$v_n = (\sqrt{\omega_1} a_{n,1}, \dots, \sqrt{\omega_R} a_{n,R}) \in \mathbb{R}^R.$$

Then $c_n = \|v_n\|_2$, and by the triangle inequality,

$$\sum_n \|v_n\|_2 \geq \left\| \sum_n v_n \right\|_2.$$

Squaring both sides gives

$$\left(\sum_n \sqrt{\sum_r \omega_r a_{n,r}^2} \right)^2 \geq \sum_r \omega_r \left(\sum_n a_{n,r} \right)^2.$$

This is exactly

$$\mathcal{Q}_{\text{shared}}^*(\theta) \geq \mathcal{Q}_{\text{group}}^*(\theta).$$

Equality in the triangle inequality holds precisely when the nonzero vectors v_n are positively collinear, giving the stated strictness condition.

B.8 Proof of Corollary 4.8

By Theorem 4.5, for each group r ,

$$Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \alpha_{t,r} Q_{r,\theta_t}(q_{r,\theta_t}^*) = \alpha_{t,r} A_r(\theta_t)^2.$$

Therefore the aggregate objective for Online Group-Specific TASS satisfies

$$\mathcal{Q}_t(\{p_{t,r}^{\text{online}}\}) = \sum_r \omega_r Q_{r,\theta_t}(p_{t,r}^{\text{online}}) \leq \sum_r \omega_r \alpha_{t,r} A_r(\theta_t)^2.$$

If this upper bound is smaller than

$$\mathcal{Q}_t(p_t^{\text{shared}}) = \sum_r \omega_r Q_{r,\theta_t}(p_t^{\text{shared}}),$$

then Online Group-Specific TASS has smaller aggregate second moment than the shared online rule.

For the comparison with the best possible shared rule, note that

$$\mathcal{Q}_{\text{group}}^*(\theta_t) = \sum_r \omega_r A_r(\theta_t)^2.$$

Thus

$$\sum_r \omega_r \alpha_{t,r} A_r(\theta_t)^2 = \bar{\alpha}_t \mathcal{Q}_{\text{group}}^*(\theta_t).$$

If

$$\bar{\alpha}_t < \frac{\mathcal{Q}_{\text{shared}}^*(\theta_t)}{\mathcal{Q}_{\text{group}}^*(\theta_t)},$$

then

$$\mathcal{Q}_t(\{p_{t,r}^{\text{online}}\}) \leq \bar{\alpha}_t \mathcal{Q}_{\text{group}}^*(\theta_t) < \mathcal{Q}_{\text{shared}}^*(\theta_t),$$

so Online Group-Specific TASS beats even the shared oracle, and therefore any feasible shared online distribution. Theorem 4.7 gives the stated lower bound and strictness interpretation for the ratio.

B.9 Optional finite-distance staleness bound

The exact staleness identity is the main result. If desired, it can be supplemented by a local Lipschitz bound showing that Static TASS remains safe close to its pilot point. Suppose for group r that $a_{n,r}(\theta)$ is Lipschitz in θ on a neighborhood U of θ_0 with constants $L_{n,r}$, that

$$\underline{A}_r = \inf_{\theta \in U} \sum_n a_{n,r}(\theta) > 0,$$

and that q_{r,θ_0}^* is bounded away from zero. Then

$$\chi^2(q_{r,\theta}^* \| q_{r,\theta_0}^*) \lesssim \|\theta - \theta_0\|_2^2.$$

This follows by writing $q_{r,\theta}^* = a_r(\theta) / \sum_n a_{n,r}(\theta)$, bounding the ℓ_1 change in the normalized vector by the ℓ_1 change in $a_r(\theta)$, and then using

$$\chi^2(u \| v) \leq \frac{\|u - v\|_1^2}{\min_n v_n}.$$

We do not rely on this bound for the main online-versus-static comparison, because the exact chi-square diagnostic is more informative and computable in the gradient-RMSE experiments.

C Code and data availability (reproducibility!)

Code for implementing our results is available at <https://github.com/diegolgg/onlinetass>. Data for our experiments was taken from <https://tardis.dev/>; specifics of data sources are available in Section 5.3. Parameter tuning is documented in the GitHub.

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